

Statistical Methods – Notes

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UNIT – I

1. Measures of Central Tendency

Definition 1.1: Measures of Central Tendency

Measures of Central Tendency are the statistical methods/measures represented by a single value that determines the central point of the distribution.

The commonly used measures of central tendency are **mean**, **median** and **mode**. They can be defined based on the ungrouped and grouped data.

1.1 Ungrouped Data

Definition 1.2: Mean (Ungrouped Data)

Ungrouped Data: The data given roughly.

Suppose we have ungrouped data $x_1, x_2, \dots, x_n$.

$$\text{Mean} = \bar{x} = \frac{\sum x_i}{n}, \quad n \rightarrow \text{no. of elements in data.}$$

Example 1.1: Mean of ungrouped data

Data: 2, 3, 5, 7, 8, 10

$$\text{Mean } \bar{x} = \frac{\sum x_i}{n} = \frac{2 + 3 + 5 + 7 + 8 + 10}{6} = 5.8$$

Definition 1.3: Median

Median: Middle value of the sorted data.

- * If data contains odd no. of elements, middle value is median.
- * If data is even no. of elements, then median is the average of the middle 2 elements.

Definition 1.4: Mode

Mode: An element which appears more no. of times in the data is known as mode. A data may have no mode / more than 1 mode.

Example 1.2: Mode of ungrouped data

2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 6 Mode $\rightarrow$ 4.

1.2 Frequency Data**Mean for Frequency Data**

x_1	x_2	$\cdots$	x_n
f_1	f_2	$\cdots$	f_n

$$\text{Mean } \bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{\sum f_i x_i}{N},$$

$$N = \sum f_i$$

Example 1.3: Mean of frequency data

Data:

x	f
2	1
3	2
4	4
5	1
6	1

$$\frac{\sum f_i x_i}{\sum f_i} = \bar{x} = \frac{1 \times 2 + 2 \times 3 + 4 \times 4 + 5 + 6}{9}$$

$$\bar{x} = \frac{2 + 6 + 16 + 11}{9} = \frac{35}{9}, \quad \bar{x} = 3.88$$

For a frequency data, median is calculated through cumulative frequency and correspondingly median lies in $\frac{N}{2}$, ($N = \sum f_i$).

Example 1.4: Median of frequency data

For the above problem, $N = 9$; $N/2 = 4.5$ and also the table becomes:

x	f	cf
2	1	1
3	2	3
4	4	7 $\leftarrow$ 4.5 Median class = 4
5	1	8
6	1	9

Based on cumulative frequency.

1.3 Grouped Data

Definition 1.5: Mean and Median for Grouped Data

For a grouped data where the values are lies in the interval with interval of difference h :

* Mean is defined by,

$$\text{Mean } \bar{x} = \frac{\sum f_i x_i}{\sum f_i} \quad \text{where } x_i \text{ is mid-point of interval.}$$

* Median is defined by,

$$l + \frac{\left(\frac{N}{2} - cf\right)}{f_i} \times h$$

and median lies in the $\frac{N}{2}$ th position, where

$l \rightarrow$ lower limit of median class

$cf \rightarrow$ cumulative frequency of the preceding median class

$f_i \rightarrow$ frequency of the median class.

Example 1.5: Mean and median of grouped data

x	f	cf	x_i
0 – 10	3	3	5
10 – 20	7	10	15
20 – 30	5	15	25

Find mean & median.

$$\text{Mean } \bar{x} = \frac{3 \times 5 + 7 \times 15 + 5 \times 25}{15} = 16.33$$

Median: $N = \sum f_i = 15$, $N/2 = 7.5$, $h = 10$.

10 – 20 $\rightarrow$ Median class.

$$10 + \left(\frac{7.5 - 3}{7}\right) \times 10 \Rightarrow 16.4$$

where 3 is the cf before median class and 7 is the median class frequency.

1.4 Mode in the Grouped Data

Mode for Grouped Data

$$\text{Mode} = l + \left(\frac{f_m - f_1}{2f_m - f_1 - f_2}\right) \times h, \quad \begin{array}{l} f_1 \rightarrow \text{Before mode class} \\ f_2 \rightarrow \text{After mode class.} \end{array}$$

High frequency represent mode class.

l – lower limit of mode class.
 f_m – Mode class frequency.

Example 1.6: Mode of grouped data

10 – 20 (Interval).

$$\text{Mode} = 10 + \left(\frac{7 - 3}{2(7) - 3 - 5} \right) \times 10$$

$$\text{Mode} = 16.66$$

2. Quartiles

Definition 2.1: Quartiles

Quartiles: Divides the whole distribution into 4 equal parts. Hence we have, 1st quartile – Q_1 :

$Q_1 \rightarrow 25\%$ percentile of data.
 $Q_2 \rightarrow 50\%$ percentile of data (Median).
 $Q_3 \rightarrow 75\%$ percentile of data.
 $Q_4 \rightarrow 100\%$ percentile of data.

2.1 Quartiles for Ungrouped Data

Quartiles – Ungrouped Data

(a) For ungrouped data Q_k can be given by,

$$Q_k = \frac{k(n+1)}{4} \text{th value, } k = 1, 2, 3.$$

$$Q_1 = \frac{n+1}{4}, \quad Q_2 = \frac{(n+1)}{2}, \quad Q_3 = \frac{3(n+1)}{4}, \quad Q_4 = (n+1)$$

(1) For even numbered data, quartiles can be get it from the average value of respective position.

Problem 2.1: Quartiles of ungrouped data (odd n)

Find the 3 quartiles Q_1, Q_2, Q_3 for the data

12, 15, 14, 10, 18, 20, 25

Solution

Sorting: 10, 12, 14, 15, 18, 20, 25

$$Q_1 = \frac{n+1}{4} = \frac{7+1}{4} = 2\text{th} = 12$$

$$Q_2 = \frac{n+1}{2} = \frac{7+1}{2} = 4\text{th} = 15$$

$$Q_3 = \frac{3(8)}{4} = \frac{24}{4} = 6\text{th} = 20$$

$$Q_4 = 8\text{th value} \rightarrow \times$$

Problem 2.2: Quartiles of ungrouped data (even positions)

Find the 3 quartiles Q_1 , Q_2 & Q_3 from data

2, 7, 8, 5, 12, 14, 21, 13, 18.

Solution

Sorted: 3, 5, 7, 8, 12, 13, 14, 18, 21

$$Q_1 = \frac{n+1}{4} = \frac{10}{4} = 2.5\text{th} = 5 \text{ \& } 7 = 6$$

$$Q_2 = \frac{n+1}{2} = \frac{10}{2} = 5\text{th} = 12$$

$$Q_3 = \frac{3(10)}{4} = \frac{15}{2} = 7.5\text{th} = 14\text{th} \text{ \& } 18 = 16$$

2.2 Quartiles for Grouped Data**Quartiles – Grouped Data**

(b) When we have grouped data in intervals, then the quartiles can be identified through cumulative frequencies where $N = \sum f_i$. Hence,

Q_1 class lies in $\frac{N}{4}$ th position.

Q_2 class lies in $\frac{N}{2}$ th position.

Q_3 class lies in $\frac{3N}{4}$ th position.

Then,

$$Q_1 = l + \frac{\left(\frac{N}{4} - cf\right)}{f_i} \times h, \quad Q_2 = l + \frac{\left(\frac{N}{2} - cf\right)}{f_i} \times h, \quad Q_3 = l + \frac{\left(\frac{3N}{4} - cf\right)}{f_i} \times h,$$

where

l = lower value of quartile class

cf = cumulative frequency before quartile class

f_i = frequency of quartile class

h = interval difference.

Problem 2.3: Quartiles of grouped data

Find all the three quartiles Q_1, Q_2, Q_3 for following data.

Class Interval	Frequency	cf
0 – 10	5	5
10 – 20	9	14
20 – 30	12	26
30 – 40	10	36
40 – 50	4	40

$$N = 40$$

Solution

$$Q_1 = \frac{N}{4}\text{th} = 10\text{th.} \quad (10 - 20 \text{ Interval}).$$

$$Q_1 = 10 + \left(\frac{10 - 5}{9} \right) \times 10 = 15.55$$

$$Q_2 = \frac{N}{2}\text{th} = 20\text{th.} \quad (20 - 30\text{th})$$

$$Q_2 = 20 + \left(\frac{20 - 14}{12} \right) \times 10 = 25$$

$$Q_3 = \frac{3(40)}{4} = 30. \quad (30 - 40 \text{ Interval})$$

$$Q_3 = 30 + \left(\frac{30 - 26}{10} \right) \times 10 = 34$$

3. Measures of Dispersion

Definition 3.1: Measures of Dispersion

Measures of Dispersion: It tells us that how the values of data spreads-out from the mean, median and quartiles.

We have following measures of dispersion:

1. Range
2. Variance

3. Standard deviation
4. Inter-quartile range
5. Mean deviation
6. Quartile deviation.

3.1 Range

Definition 3.2: Range

(A) **Range:** The difference between max & min value of data.

$$\text{Range} = \text{Max} - \text{Min}$$

Example 3.1: Range

For the data 4, 7, 9, 10, 15, 18:

$$\text{range} = 14$$

3.2 Variance

Definition 3.3: Variance (Ungrouped Data)

(B) **Variance:** For the ungrouped data defined by variation

$$\sigma^2 = \frac{1}{n} \sum (x_i - \bar{x})^2 \quad (\text{or}) \quad \sigma^2 = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2$$

Example 3.2: Variance and standard deviation

3, 6, 8, 10, 12, $n = 5$, $\bar{x} = 7.8$

$$\sigma^2 = 70.6 - (60.84) = 9.76$$

$$\boxed{\sigma = 3.12} \rightarrow \text{standard deviation}$$

Note

Standard deviation is the square root of σ^2 .

Definition 3.4: Variance (Grouped Data)

Variance for grouped data defined by

$$\sigma^2 = \frac{1}{N} \sum f_i(x_i - \bar{x})^2 \quad (\text{or}) \quad \sigma^2 = \frac{\sum f_i x_i^2}{N} - \left(\frac{\sum f_i x_i}{N} \right)^2$$

In the case of interval data x_i is the midpoint.

Example 3.3: Variance of grouped data

Data,

Class Interval	Frequency	Mid-point x_i	x_i^2
0 – 10	3	5	25
10 – 20	7	15	225
20 – 30	12	25	625
30 – 40	8	35	1225

$$\sigma^2 = \frac{18950}{30} - \left(\frac{700}{30} \right)^2$$

$$\sigma^2 = 631.66 - 544.44$$

$$\sigma^2 = 87.22$$

$$\sigma = \sqrt{87.22} = 9.3$$

$$\text{Mean } \bar{x} = \frac{\sum f_i x_i}{N} = 23.33$$

3.3 Inter-quartile Range**Problem 3.1: Inter-quartile range**

Calculate interquartile range for the data,

class	frequency	cf
0 – 10	5	5
10 – 20	8	13 $\rightarrow Q_1$
20 – 30	12	25
30 – 40	10	35 $\rightarrow Q_3$

$$N = 35$$

Solution

$$I_{qr} = Q_3 - Q_1$$

$$Q_1 = \frac{N}{4} \text{th} = \frac{35}{4} = 8.75 \text{th} = (10 \text{ to } 20)$$

$$Q_1 = 10 + \left(\frac{8.75 - 5}{8} \right) \times 10 = 14.68 \quad (8 \rightarrow f_i)$$

$$Q_3 = \frac{3N}{4} \text{th} = \frac{3(35)}{4} = 26.25 \text{th} = (30 \text{ to } 40)$$

$$Q_3 = 30 + \left(\frac{26.25 - 25}{10} \right) \times 10 = 31.25 \quad (10 \rightarrow f_i)$$

$$I_{qr} = 31.25 - 14.68$$

$$I_{qr} = 16.57$$

3.4 Mean Deviation

Definition 3.5: Mean Deviation

(E) Mean Deviation:

$$(M.D)_{\text{mean}} = \frac{1}{n} \sum |x_i - \bar{x}|$$

For grouped:

$$(M.D)_{\text{mean}} = \frac{1}{N} \sum f_i |x_i - \bar{x}|$$

Mean deviation about median:

$$(M.D)_{\text{median}} = \frac{1}{n} \sum |x_i - \text{median}|$$

For grouped:

$$(M.D)_{\text{median}} = \frac{1}{N} \sum f_i |x_i - \text{median}|$$

3.5 Quartile Deviation

Definition 3.6: Quartile Deviation

It is defined by QD :

$$QD = \frac{Q_3 - Q_1}{2}$$

3.6 Co-efficient of Variation (COV)

Definition 3.7: Co-efficient of Variation

Tells us that at what percentage level the variation present in the data, and is defined by CV :

$$CV = \frac{\sigma}{\bar{x}} \times 100 = \frac{S.D}{\text{Mean}} \times 100$$

3.7 Problems

Problem 3.2: Average profit and coefficient of variation

A small town a survey was conducted in respect of the income made by retails of shops. The following results were obtained.

Profit (In 1000 Rs)	No. of Shops f_i	x_i	$f_i x_i^2$
-4 to -3	4	-3.5	49
-3 to -2	10	-2.5	62.5
-2 to -1	22	-1.5	49.5
-1 to 0	28	-0.5	7
0 to 1	38	0.5	9.5
1 - 2	56	1.5	126
2 - 3	40	2.5	250
3 - 4	24	3.5	294
4 - 5	18	4.5	364.5
5 - 6	10	5.5	302.5

Find the average profit, total profit by all shops and co-efficient of variation of earnings.

Solution

From data, $N = 250$.

$$\text{Mean } \bar{x} = \frac{\sum f_i x_i}{N} = \frac{337}{250} = 1.348$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

$$\sigma^2 = \frac{1}{N} \sum f_i x_i^2 - \left(\frac{\sum f_i x_i}{N} \right)^2$$

$$\sigma^2 = \frac{1514.5}{250} - \left(\frac{337}{250} \right)^2 = 6.058 - 1.817$$

$$\sigma^2 = 4.2409$$

$$\sigma = 2.05$$

$$Cov = \frac{2.05}{1.348} \times 100 = 152.077$$

$$\text{Total profit} = N \times \bar{x} = 250 \times 1.348 = \underline{337}$$

Problem 3.3: Comparing variability of two companies

A purchasing agent receives samples of airbags from 2 companies. He has the samples tested in his own labs for testing weights and has arrived at the following results.

Testing range (grms)	Company A	Company B	x_i	$f_i x_{iA}^2$	$f_i x_{iB}^2$
50 – 60	3	10	55	331950	358150
60 – 70	42	16	65		
70 – 80	22	36	75		
80 – 90	3	8	85		
	$N = 70$	$N = 70$			

Which company's airbags show more variable in quality?

Solution

For A:

$$\bar{x}_A = \frac{\sum f_i x_i}{N} = \frac{4800}{70} = 68.57$$

$$\sigma^2 = \frac{331950}{70} - \left(\frac{4800}{70} \right)^2 = 4742.14 - 4702$$

$$\sigma^2 = 40.1$$

$$\sigma_A = 6.33$$

$$CV = \frac{6.33}{68.57} \times 100 = 9.23$$

For B:

$$\bar{x}_B = \frac{\sum f_i x_i}{70} = \frac{4970}{70} = 71$$

$$\sigma^2 = \frac{358150}{70} - \left(\frac{4970}{70} \right)^2 = 5116.42 - 5041$$

$$\sigma^2 = 75.42$$

$$\sigma_B = 8.68$$

$$C.V = \frac{8.68}{71} \times 100 = 12.22$$

Problem 3.4: Mean deviation and quartile deviation

For the data, find mean deviation about mean, median and quartile deviation.

Class Interval	No. of items	x_i	cf
0 – 10	10	5	10
10 – 20	15	15	25
20 – 30	25	25	50
30 – 40	10	35	60

$$N = 60$$

Solution

$$\bar{x} = \frac{1250}{60} = 20.83$$

$$(M.D)_{\text{mean}} = \frac{1}{N} \sum f_i |x_i - \bar{x}|$$

$$(M.D)_{\text{median}} = \frac{1}{N} \sum f_i |x_i - \text{Median}|$$

$$\begin{aligned}(M.D)_{\text{mean}} &= \frac{1}{60} [10 |5 - 20.83| + 15 |15 - 20.83| + 25 |25 - 20.83| + 10 |35 - 20.83|] \\&= \frac{1}{60} [158.3 + 87.45 + 104.25 + 141.7] \\(M.D)_{\text{mean}} &= \underline{8.195}\end{aligned}$$

Median:

$$l + \frac{\left(\frac{N}{2} - cf\right)}{f_i} \times h, \quad \frac{N}{2} = 30$$

$$20 - 30: \quad f_i = 25, \quad cf = 25.$$

$$20 + \left(\frac{30 - 25}{25}\right) \times 10 \Rightarrow 22$$

$$\begin{aligned}(M.D)_{\text{median}} &= \frac{1}{N} \sum f_i |x_i - \text{median}| \\&= \frac{1}{60} [10 |5 - 22| + 15 |15 - 22| + 25 |25 - 22| + 10 |35 - 22|] \\&\Rightarrow \frac{1}{60} [170 + 105 + 75 + 130] = \underline{8}\end{aligned}$$

Quartile deviation $\frac{Q_3 - Q_1}{2}$:

$$Q_1 = \frac{N}{4} = 15\text{th} \quad (10 - 20), \quad f_i = 15, \quad cf = 10$$

$$Q_1 = 10 + \left(\frac{15 - 10}{15}\right) \times 10 = 13.33$$

$$Q_3 = \frac{3N}{4} = \frac{3(60)}{4} = 45\text{th} \quad (20 - 30), \quad f_i = 25, \quad cf = 25$$

$$Q_3 = 20 + \left(\frac{45 - 25}{25}\right) \times 10 = 28$$

$$\frac{Q_3 - Q_1}{2} = \underline{7.335}$$

4. Moments, Skewness and Kurtosis

4.1 Concepts of Moments

Definition 4.1: Concepts of Moments

Concepts of Moments: Quantitative measures used to describe the shape of distribution. The r th moment about a point gives insight into the distribution's characteristics.

Moments: Measures of shape of a set of points.

Skewness: Measure of asymmetry of probability distribution.

Kurtosis: Measure of “tailedness” of probability distribution.

4.2 Raw and Central Moments

Definition 4.2: Raw and Central Moments

* **Raw moment (about origin):**

$$\mu'_r = \frac{1}{n} \sum_{i=1}^n x_i^r$$

From the above $\mu'_1 = \bar{x}$ (mean).

* **Central moment (about mean):**

$$\mu_r = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^r, \quad \boxed{\mu_1 = 0}$$

From above, $\mu_1 = 0$;

$$\mu_2 (r = 2) = \sigma^2 \text{ (variance of } x \text{)}$$

Note

1. Third central moment μ_3 measures skewness.
2. 4th central moment μ_4 measures kurtosis.

4.3 Relation between Raw & Central Moments

Raw $\leftrightarrow$ Central Moments

Central moments can be expressed in terms of raw moments:

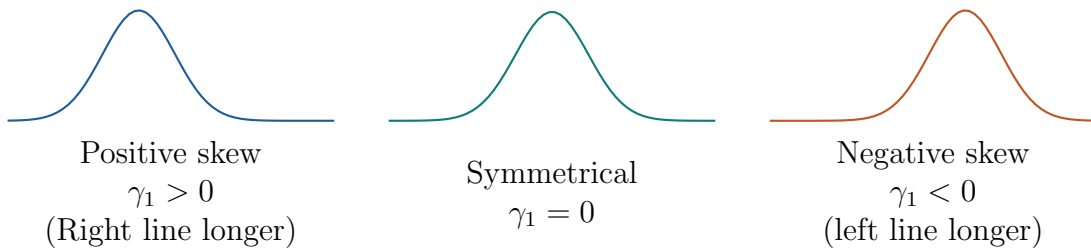
$$\begin{aligned}\mu_1 &= 0 \\ \mu_2 &= \mu'_2 - (\mu'_1)^2 \\ \mu_3 &= \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3 \\ \mu_4 &= \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4\end{aligned}$$

4.4 Skewness**Definition 4.3: Skewness**

Skewness: Measure of the asymmetry of probability of distribution.

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} \quad (\text{or}) \quad \gamma_1 = \sqrt{\beta_1} = \frac{\mu_3}{\mu_2^{3/2}}$$

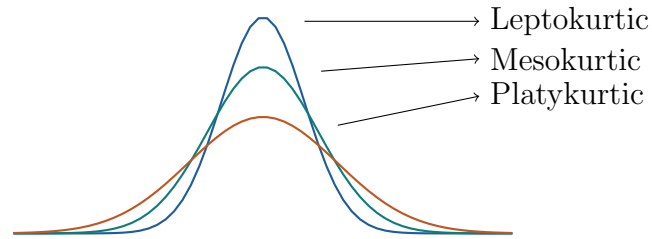
Visual representation:

**4.5 Kurtosis****Definition 4.4: Kurtosis**

$$\beta_2 = \frac{\mu_4}{\mu_2^2} \quad (\text{or}) \quad \gamma_2 = \beta_2 - 3$$

Types:

- Mesokurtic:** general normal distribution ($\beta_2 = 3$)
- Leptokurtic:** More peaked than normal ($\beta_2 > 3$)
- Platykurtic:** Less peaked than normal ($\beta_2 < 3$).



4.6 Interpretation of β_1 and β_2

Note

Skewness (β_1):

- $\beta_1 = 0 \rightarrow$ perfectly symmetrical.
- Sign of μ_3 gives direction: $\mu_3 > 0$ right skewed, $\mu_3 < 0$ left skewed.

Kurtosis (β_2):

- * $\beta_2 = 3 \rightarrow$ mesokurtic
- * $\beta_2 > 3 \rightarrow$ leptokurtic
- * $\beta_2 < 3 \rightarrow$ platykurtic.

4.7 Problems

Problem 4.1: Skewness and kurtosis of ungrouped data

For data, 3, 7, 7, 2, 9 ($n = 5$). Skewness & kurtosis?

Solution

Mean $\bar{x} = 5.6$.

$$\mu_r = \frac{1}{n} \sum (x_i - \bar{x})^r, \quad \bar{x} = 5.6$$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$(x_i - \bar{x})^3$	$(x_i - \bar{x})^4$
3	-2.6	6.76	-17.57	45.69
7	+1.4	1.96	2.744	3.84
7	+1.4	1.96	2.744	3.84
2	-3.6	12.96	-46.65	167.96
9	3.4	11.56	39.304	133.63
Σ :	0	35.2	-19.428	354.96

$$\mu_1 = 0$$

$$\mu_2 = \frac{1}{n} \sum (x_i - \bar{x})^2 = \frac{35.2}{5} = 7.04$$

$$\mu_3 = \frac{1}{n}(-19.428) = -3.88$$

$$\mu_4 = 70.992$$

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{(-3.88)^2}{(7.04)^3} = \frac{15.05}{348.91} = 0.043$$

$$\boxed{\beta_1 = 0.043} \rightarrow \text{Skewness} \quad \boxed{\gamma_1 = -0.207} \text{ slightly negatively skewed.}$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{70.992}{(7.04)^2} = 1.432$$

$$\gamma_2 = \beta_2 - 3 = 1.432 - 3 = -1.568 < 3 \rightarrow \text{platykurtic.}$$

Problem 4.2: Skewness and kurtosis of grouped data

For given data, find skewness & kurtosis.

Class Int	Frequency	x_i	$(x_i - \bar{x})$	$(x_i - \bar{x})^2$	$(x_i - \bar{x})^3$	$(x_i - \bar{x})^4$
0 – 10	5	5	-22	+484	-10648	+234256
10 – 20	8	15	-12	144	-1728	20736
20 – 30	15	25	-2	4	-8	16
30 – 40	16	35	8	64	512	4096
40 – 50	6	45	18	324	5832	104976

$N = 50$

Solution

$$\bar{x} = \frac{\sum f_i x_i}{N} = \frac{1350}{50} = 27$$

fd^2	fd^3	fd^4
2420	-53240	-1171280
1152	-13824	165888
60	-120	240
1024	8192	65536
1944	34992	629856

$$\sum fd^2 = 6600 \quad (-24000)$$

$$\sum fd^3 = 82480$$

$$\sum fd^4 = -309760$$

$$\mu_1 = 0$$

$$\mu_2 = \frac{1}{n}(2244) = 44.88$$

$$\mu_3 = \frac{1}{n}(82480) = 1649.6$$

$$\mu_4 = \frac{1}{n}(-309760) =$$

4.8 Bowley's and Karl–Pearson's Coefficients of Skewness

Definition 4.5: Bowley's and Karl–Pearson's Coefficients

Other than the definition of moments using mean, median, quartiles, we can identify the skewness, respectively they are called,

1. Bowley's Co-efficient of Skewness
2. Karl–Pearson's of skewness.

Bowley's Co-efficient of Skewness is defined by,

$$S_{KB} = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

Karl–Pearson's:

$$S_{KP} = \frac{3(\text{Mean} - \text{Median})}{\sigma}$$

Note

- $S_{KB} \rightarrow$ Midpoint is not necessary.
- If the given data is open-ended data (below 20) / (above 70) & in such cases S_{KB} is a perfect fit for calculating skewness.

Problem 4.3: Bowley's coefficient of skewness

Calculate Bowley's coefficient of skewness, from data,

Profits (in Lakhs)	No. of Items (<i>cf</i>)
< 10	8
< 20	20
< 30	40
< 40	50
< 50	56
< 60	59
< 70	60

Cl. Profit	f_i
< 10	8
10 – 20	12
20 – 30	20
30 – 40	10
40 – 50	6
50 – 60	3
60 – 70	1
	60

$N = 60$

Solution

$$Q_1 = \frac{N}{4} = 15, \quad Q_2 = 30, \quad Q_3 = 45$$

$$Q_1 = 15, \quad Q_2 = 30, \quad Q_3 = 45 \text{ (positions)}$$

$$Q_1 = (10 \text{ to } 20), \quad l = 10, \quad f_i = 12, \quad N = 60, \quad cf = 8$$

$$Q_1 = 10 + \left(\frac{15 - 8}{12} \right) \times 10 = 15.83$$

$$Q_2 = 20 + \left(\frac{30 - 20}{20} \right) \times 10 = 25$$

$$Q_3 = (30 - 40) = 30 + \left(\frac{45 - 40}{10} \right) \times 10 = 35$$

$$S_{KB} = \frac{35 + 15.83 - 50}{35 - 15.83} = \underline{0.043}$$

Problem 4.4: Karl–Pearson’s coefficient of skewness

Calculate Karl–Pearson’s coefficient of skewness for data,

Profit (in lakhs)	No. of Companies	cf	x_i	$f_i x_i$	$f_i x_i^2$
10 – 20	18	18	15	270	4050
20 – 30	20	38	25	500	12500
30 – 40	30	68	35	1050	36750
40 – 50	22	90	45	990	44550
50 – 60	10	100	55	550	30250
	$N = 100$			3360	128100

Solution

$$Q_2 = \frac{N}{2} = 50 = 30 + \left(\frac{50 - 38}{20} \right) \times 10 = \underline{34}$$

$$\text{Mean } \bar{x} = \frac{\sum f_i x_i}{N} = \frac{3360}{100} = \underline{33.60}$$

$$\sigma^2 = \frac{\sum f_i x_i^2}{N} - \left(\frac{\sum f_i x_i}{N} \right)^2$$

$$= \frac{128100}{100} - \left(\frac{3360}{100} \right)^2 = 1281 - 1128.96$$

$$\sigma^2 = 152.04$$

$$\sigma = 12.33$$

$$S_{KP} = \frac{3(33.60 - 34)}{12.33} = \underline{-0.097}$$

Statistical Methods

Probability and Random Processes — Unit I

Random Variables, Statistical Averages, Moment Generating Functions,
Binomial, Poisson and Normal Distributions

Consolidated Theory, Results and Fully Worked Problems

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These consolidated notes bring together the five lecture modules of Unit I: **Random Variables, Statistical Averages, Moment Generating Functions, Binomial and Poisson Distributions** and the **Normal Distribution** — with complete theory, standard results and fully worked solutions.

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1 Random Variables

1.1 What Do We Mean by a Random Variable?

- Experiments often have *non-numerical* outcomes.
- Example:** a box has 3 red, 5 black and 2 white chips; 2 items are drawn. The pair may be (i) both white, (ii) both red, (iii) both black, (iv) one white and one red, (v) one white and one black, (vi) one red and one black.
- We *assign real numbers* to such outcomes.

Definition — Random Variable

Assigning a real number to every outcome of a random experiment gives a **random variable**, denoted $X, Y, Z, \dots$ or $X_1, X_2, X_3, \dots$

In the example above, if X = number of white chips drawn, then X takes the values 0, 1, 2.

1.2 Classification

Definition — Discrete Random Variable (DRV)

Assumes a discrete set of values.

Example: the random variable X discussed above.

Definition — Continuous Random Variable (CRV)

Assumes values in an interval.

Example: the life length of an electrical good.

1.3 Probability Function (PDF)

DRV — Probability Distribution (mass) Function

In tabular form:

$X = x$	x_1	x_2	$\dots$	x_n
$P(X = x)$	$P(X = x_1)$	$P(X = x_2)$	$\dots$	$P(X = x_n)$

CRV — Probability Density Function

A curve, or a functional form, for instance

$$f(x) = e^{-x}, \quad x \geq 0.$$

1.4 Properties of a Probability Density Function

If X is a DRV

- (i) $f(x) \geq 0$
- (ii) $\sum f(x) = 1$

If X is a CRV

- (i) $f(x) \geq 0$
- (ii) $\int_{-\infty}^{\infty} f(x) dx = 1$

1.5 Cumulative Distribution Function

$$F(x) = P(X \leq x)$$

CDF from PDF

$$F(x) = \begin{cases} \sum_{x_i \leq x} P(X = x_i), & X \text{ a DRV} \\ \int_{-\infty}^x f(x) dx, & X \text{ a CRV} \end{cases}$$

PDF from CDF

$$f(x) = \begin{cases} F(x_i) - F(x_{i-1}), & X \text{ a DRV} \\ \frac{d}{dx} F(x), & X \text{ a CRV} \end{cases}$$

1.6 Worked Problems

Problem 1.1 — Probability Distribution of White Chips

A box contains 3 red and 2 white coloured chips and 2 chips are drawn. If X denotes the number of white chips drawn, determine the PDF of X .

Solution — Setting up

X = number of white chips drawn, so X takes the values

$$X = 0 \text{ (no white),} \quad X = 1 \text{ (one white),} \quad X = 2 \text{ (two white).}$$

The total number of ways of drawing 2 chips from 5 is ${}^5C_2 = 10$.

Solution — The probabilities

$$P(X = 0) = P(\text{both are red}) = \frac{{}^3C_2}{{}^5C_2} = \frac{3}{10},$$

$$P(X = 1) = P(\text{one white and one red}) = \frac{{}^2C_1 \times {}^3C_1}{{}^5C_2} = \frac{6}{10},$$

$$P(X = 2) = P(\text{both are white}) = \frac{{}^2C_2}{{}^5C_2} = \frac{1}{10}.$$

Solution — PDF of X (X is a DRV)

$X = x$	0	1	2
$P(X = x)$	$\frac{3}{10}$	$\frac{6}{10}$	$\frac{1}{10}$

$$\text{Check: } \frac{3}{10} + \frac{6}{10} + \frac{1}{10} = 1. \checkmark$$

Problem 1.2 — Time Spent on an Electronic Gadget

The length of time (in minutes) that a person spends on an electronic gadget is a random phenomenon with probability function

$$f(x) = \begin{cases} A e^{-x/5}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

- (i) Find the value of A .
- (ii) What is the probability that the person spends (a) more than 10 minutes, (b) less than 5 minutes, (c) between 5 and 10 minutes?

Solution — (i) The value of A

Since $f(x)$ is a pdf, it satisfies $\int_{-\infty}^{\infty} f(x) dx = 1$:

$$\begin{aligned} \left(\int_{-\infty}^0 + \int_0^{\infty} \right) f(x) dx &= 1 \\ 0 + \int_0^{\infty} A e^{-x/5} dx &= 1 \implies A \left[\frac{e^{-x/5}}{-1/5} \right]_0^{\infty} = 1 \\ -5A(0 - 1) &= 1 \end{aligned}$$

$$A = \frac{1}{5}$$

$$\text{Hence } f(x) = \begin{cases} \frac{1}{5} e^{-x/5}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

Solution — (ii)(a) more than 10 minutes

$$\begin{aligned} P(X > 10) &= \int_{10}^{\infty} \frac{1}{5} e^{-x/5} dx \\ &= \frac{1}{5} \left[\frac{e^{-x/5}}{-1/5} \right]_{10}^{\infty} \\ &= - \left(0 - e^{-2} \right) = \boxed{\frac{1}{e^2}} \end{aligned}$$

Solution — (ii)(b) less than 5 minutes

$$\begin{aligned} P(X < 5) &= \int_0^5 \frac{1}{5} e^{-x/5} dx \\ &= - \left(e^{-1} - 1 \right) \\ &= \boxed{1 - \frac{1}{e}} \end{aligned}$$

Solution — (ii)(c) between 5 and 10 minutes

$$\begin{aligned}
 P(5 < X < 10) &= \int_5^{10} f(x) dx = \int_5^{10} \frac{1}{5} e^{-x/5} dx \\
 &= -\left(e^{-2} - e^{-1}\right) = e^{-1} - e^{-2} = \frac{1}{e} - \frac{1}{e^2} = \boxed{\frac{e-1}{e^2}}
 \end{aligned}$$

Problem 1.3 — Determining a and b for a Given PDF

A continuous random variable X has the pdf $f(x) = 3x^2$, $0 \leq x \leq 1$. Find a and b such that

- (i) $P(X \leq a) = P(X > a)$,
- (ii) $P(X > b) = 0.05$.

Solution — (i) Finding a

$$\begin{aligned}
 P(X \leq a) &= P(X > a) \\
 \int_0^a f(x) dx &= \int_a^1 f(x) dx \\
 \int_0^a 3x^2 dx &= \int_a^1 3x^2 dx \\
 3 \left[\frac{x^3}{3} \right]_0^a &= 3 \left[\frac{x^3}{3} \right]_a^1 \\
 a^3 &= 1 - a^3 \implies 2a^3 = 1
 \end{aligned}$$

$$a = \left(\frac{1}{2}\right)^{1/3} = (0.5)^{1/3}$$

Solution — (ii) Finding b

$$\begin{aligned}
 P(X > b) &= 0.05 \\
 \int_b^1 f(x) dx &= 0.05 \implies \int_b^1 3x^2 dx = 0.05 \\
 \left[x^3 \right]_b^1 &= 0.05 \implies 1 - b^3 = 0.05 \\
 b^3 &= 0.95
 \end{aligned}$$

$$b = (0.95)^{1/3}$$

Answers: $a = (0.5)^{1/3}$, $b = (0.95)^{1/3}$.

Problem 1.4 — Discrete PDF Involving a Parameter k

A random variable X has the PDF

X	0	1	2	3	4	5	6	7
$f(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

Find (i) the value of k ; (ii) $P(X < 6)$; (iii) $P(X \geq 6)$; (iv) $P(0 < X < 6)$; (v) $P(X \geq 5 | X > 3)$; (vi) the CDF of X .

Solution — (i) The value of k

X is a DRV, so $\sum f(x) = 1$:

$$k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$$

$$10k^2 + 9k = 1 \implies 10k^2 + 9k - 1 = 0$$

$$10k^2 + 10k - k - 1 = 0 \implies 10k(k + 1) - (k + 1) = 0$$

$$(10k - 1)(k + 1) = 0 \implies k = -1 \text{ or } k = \frac{1}{10}$$

Since $f(x) \geq 0$, k cannot be negative.

$$k = \frac{1}{10}$$

Solution — The PDF of X with $k = 1/10$

$X = x$	0	1	2	3	4	5	6	7
$P(X = x)$	0	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{1}{100}$	$\frac{2}{100}$	$\frac{17}{100}$

$$(\text{Here } 7k^2 + k = \frac{7}{100} + \frac{10}{100} = \frac{17}{100}.)$$

Solution — (ii), (iii) and (iv)

$$(ii) P(X < 6) = 1 - P(X \geq 6) = 1 - (P(X = 6) + P(X = 7)) = 1 - \frac{19}{100} = \boxed{\frac{81}{100}}$$

$$(iii) P(X \geq 6) = P(X = 6) + P(X = 7) = \frac{2}{100} + \frac{17}{100} = \boxed{\frac{19}{100}}$$

$$(iv) P(0 < X < 6) = P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) \\ = \frac{8}{10} + \frac{1}{100} = \boxed{\frac{81}{100}}$$

Solution — (v) Conditional probability

Using $P(A | B) = \frac{P(A \cap B)}{P(B)}$ with $(X \geq 5) \cap (X > 3) = (X \geq 5)$:

$$\begin{aligned} P(X \geq 5 | X > 3) &= \frac{P(X \geq 5)}{P(X > 3)} = \frac{P(X = 5) + P(X = 6) + P(X = 7)}{1 - P(X \leq 3)} \\ &= \frac{\frac{1}{100} + \frac{2}{100} + \frac{17}{100}}{1 - \frac{5}{10}} = \frac{20/100}{1/2} = \frac{1/5}{1/2} = \boxed{\frac{2}{5}} \end{aligned}$$

Solution — (vi) CDF of X , $F(x) = P(X \leq x)$

$X = x$	Computation	$F(x) = P(X \leq x)$
0	0	0
1	$0 + \frac{1}{10}$	$\frac{1}{10}$
2	$\frac{1}{10} + \frac{2}{10}$	$\frac{3}{10}$
3	$\frac{3}{10} + \frac{2}{10}$	$\frac{5}{10}$
4	$\frac{5}{10} + \frac{3}{10}$	$\frac{8}{10}$
5	$\frac{8}{10} + \frac{1}{100}$	$\frac{81}{100}$
6	$\frac{81}{100} + \frac{2}{100}$	$\frac{83}{100}$
7	$\frac{83}{100} + \frac{17}{100}$	1

Problem 1.5 — PDF from a Given CDF, and Back Again

The CDF of a continuous random variable X is

$$F(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \leq x \leq \frac{1}{2} \\ 1 - \frac{3}{25}(3-x)^2, & \frac{1}{2} \leq x < 3 \\ 1, & x \geq 3. \end{cases}$$

Determine the PDF of X , and conversely recover the CDF from it.

Solution — PDF from the CDF

For a CRV, $f(x) = \frac{d}{dx}F(x)$. Differentiating piecewise (note $\frac{d}{dx}[-\frac{3}{25}(3-x)^2] = \frac{6}{25}(3-x)$):

$$f(x) = \begin{cases} 0, & x < 0 \\ 2x, & 0 \leq x \leq \frac{1}{2} \\ \frac{6}{25}(3-x), & \frac{1}{2} \leq x < 3 \\ 0, & x \geq 3. \end{cases}$$

Solution — CDF from the PDF — cases $x < 0$ and $0 \leq x \leq \frac{1}{2}$

For $x < 0$:

$$F(x) = \int_{-\infty}^x 0 \, dx = 0.$$

For $0 \leq x \leq \frac{1}{2}$:

$$F(x) = \int_{-\infty}^0 0 \, dx + \int_0^x 2x \, dx = 2 \left[\frac{x^2}{2} \right]_0^x = x^2.$$

Solution — CDF from the PDF — case $\frac{1}{2} \leq x < 3$

$$\begin{aligned} F(x) &= \int_{-\infty}^0 0 \, dx + \int_0^{1/2} 2x \, dx + \int_{1/2}^x \frac{6}{25} (3-x) \, dx \\ &= \left[x^2 \right]_0^{1/2} + \frac{6}{25} \left[\frac{(3-x)^2}{-2} \right]_{1/2}^x \\ &= \frac{1}{4} - \frac{3}{25} \left[(3-x)^2 - \left(\frac{5}{2} \right)^2 \right] \\ &= \frac{1}{4} + \frac{3}{4} - \frac{3}{25} (3-x)^2 = 1 - \frac{3}{25} (3-x)^2. \end{aligned}$$

Solution — CDF from the PDF — case $x \geq 3$

$$\begin{aligned} F(x) &= \int_{-\infty}^0 0 \, dx + \int_0^{1/2} 2x \, dx + \int_{1/2}^3 \frac{6}{25} (3-x) \, dx + \int_3^x 0 \, dx \\ &= \frac{1}{4} + \frac{6}{25} \left[\frac{(3-x)^2}{-2} \right]_{1/2}^3 = \frac{1}{4} - \frac{3}{25} \left[0^2 - \left(\frac{5}{2} \right)^2 \right] \\ &= \frac{1}{4} - \frac{3}{25} \left(-\frac{25}{4} \right) = \frac{1}{4} + \frac{3}{4} = 1. \end{aligned}$$

Hence the obtained CDF and PDF are consistent. ✓

Problem 1.6 — PDF and CDF from Proportionality Conditions

A random variable X takes the values 1, 2, 3, 4 such that

$$2P(X=1) = 3P(X=2) = P(X=3) = 5P(X=4).$$

Find the PDF and the CDF of X .

Solution — Finding the common value k

Let the common value be k , i.e. $2P(X=1) = 3P(X=2) = P(X=3) = 5P(X=4) = k$. Then

$$P(X=1) = \frac{k}{2}, \quad P(X=2) = \frac{k}{3}, \quad P(X=3) = k, \quad P(X=4) = \frac{k}{5}.$$

Since X is a DRV, $\sum f(x) = 1$:

$$\frac{k}{2} + \frac{k}{3} + k + \frac{k}{5} = 1 \implies \left(\frac{15 + 10 + 30 + 6}{30} \right) k = 1$$

$$\frac{61k}{30} = 1$$

$$k = \frac{30}{61}$$

Solution — PDF and CDF of X

$X = x$	1	2	3	4
$P(X = x)$	$\frac{15}{61}$	$\frac{10}{61}$	$\frac{30}{61}$	$\frac{6}{61}$
$F(x) = P(X \leq x)$	$\frac{15}{61}$	$\frac{25}{61}$	$\frac{55}{61}$	1

Problem 1.7 — Verifying that a Function is a PDF

Show that

$$f(x) = \begin{cases} x e^{-x^2/2}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

is a probability density function.

To prove

Since x takes values in an interval, X is a CRV, so we must show (i) $f(x) \geq 0$ and (ii) $\int_{-\infty}^{\infty} f(x) dx = 1$.

Solution

(i) Clearly $f(x) = x e^{-x^2/2} \geq 0$ for all $x \geq 0$.

(ii)

$$\int_{-\infty}^{\infty} f(x) dx = \left(\int_{-\infty}^0 + \int_0^{\infty} \right) f(x) dx = 0 + \int_0^{\infty} x e^{-x^2/2} dx.$$

Put $\frac{x^2}{2} = t \implies \frac{2x dx}{2} = dt \implies x dx = dt$ (no change in the limits):

$$\int_0^{\infty} e^{-t} dt = \left[\frac{e^{-t}}{-1} \right]_0^{\infty} = -(0 - 1) = 1.$$

By (i) and (ii), $f(x)$ is a pdf of a CRV X . ■

1.7 Additional Problems — Bayes' Theorem

Problem 1.8 — All Items in the Bag are White

A bag contains 5 items and is equally likely to contain 2, 3, 4 or 5 white items. Two items are drawn at random and both turn out to be white. What is the probability that *all* the items in the bag are white?

Solution — The events

A_1 : the bag contains 2 white items

A_2 : the bag contains 3 white items

A_3 : the bag contains 4 white items

A_4 : the bag contains 5 white items

E : drawing of 2 white items

with $P(A_1) = P(A_2) = P(A_3) = P(A_4) = \frac{1}{4}$.

Solution — Conditional probabilities

$$P(E | A_1) = \frac{{}^2C_2}{{}^5C_2} = \frac{1}{10},$$

$$P(E | A_2) = \frac{{}^3C_2}{{}^5C_2} = \frac{3}{10},$$

$$P(E | A_3) = \frac{{}^4C_2}{{}^5C_2} = \frac{6}{10},$$

$$P(E | A_4) = \frac{{}^5C_2}{{}^5C_2} = 1.$$

Solution — Applying Bayes' theorem

$$\begin{aligned} P(\text{all the items in the bag are white}) &= P(A_4 | E) \\ &= \frac{P(A_4) P(E | A_4)}{P(A_1)P(E | A_1) + P(A_2)P(E | A_2) + P(A_3)P(E | A_3) + P(A_4)P(E | A_4)} \\ &= \frac{\frac{1}{4}(1)}{\frac{1}{4}\left(\frac{1}{10}\right) + \frac{1}{4}\left(\frac{3}{10}\right) + \frac{1}{4}\left(\frac{6}{10}\right) + \frac{1}{4}(1)} \end{aligned}$$

$$P(A_4 | E) = \frac{1}{2} = 50\%$$

Problem 1.9 — Liver Disease and Regular Drinkers

In a certain town, 25% of the people drink alcohol regularly. The probability of death due to liver disease is 6 times greater for a regular drinker than for a non-drinker. If the probability of death due to liver disease in the town is 0.005, what is the probability that a person who died of liver disease was a regular drinker?

Solution — The events

A_1 : person drinks regularly, $P(A_1) = 0.25$

A_2 : person does not drink regularly, $P(A_2) = 0.75$

E : death due to liver disease, $P(E) = 0.005$

Solution — Solution

Let $P(E | A_1) = 6 P(E | A_2) = k$, so $P(E | A_2) = \frac{k}{6}$. From the total probability rule,

$$P(E) = P(A_1)P(E | A_1) + P(A_2)P(E | A_2)$$

$$0.005 = 0.25k + 0.75 \cdot \frac{k}{6} = 0.375k \implies k = \frac{1}{75}$$

By Bayes' theorem,

$$P(A_1 | E) = \frac{P(A_1)P(E | A_1)}{P(E)} = \frac{0.25 \times \frac{1}{75}}{0.005}$$

$$P(A_1 | E) = \frac{2}{3}$$

Problem 1.10 — All Items in the Lot are Good

A lot of 5 items is equally likely to contain 0, 1, 2 or 3 defective items. Two items are drawn at random and both are found to be good. Find the probability that all the items in the lot are good.

Solution — The events

A_1 : 0 defective, 5 good $P(A_1) = \frac{1}{4}$

A_2 : 1 defective, 4 good $P(A_2) = \frac{1}{4}$

A_3 : 2 defective, 3 good $P(A_3) = \frac{1}{4}$

A_4 : 3 defective, 2 good $P(A_4) = \frac{1}{4}$

E : drawing of 2 good items

Solution — Conditional probabilities

$$P(E | A_1) = \frac{{}^5C_2}{{}^5C_2} = 1,$$

$$P(E | A_2) = \frac{{}^4C_2}{{}^5C_2} = \frac{6}{10},$$

$$P(E | A_3) = \frac{{}^3C_2}{{}^5C_2} = \frac{3}{10},$$

$$P(E | A_4) = \frac{{}^2C_2}{{}^5C_2} = \frac{1}{10}.$$

Solution — Applying Bayes' theorem

Given that E has happened,

$$\begin{aligned} P(\text{all the items in the lot are good}) &= P(A_1 | E) = \frac{P(A_1) P(E | A_1)}{\sum_{i=1}^4 P(A_i) P(E | A_i)} \\ &= \frac{\frac{1}{4}(1)}{\frac{1}{4}\left(1 + \frac{6}{10} + \frac{3}{10} + \frac{1}{10}\right)} \end{aligned}$$

$$P(A_1 | E) = \frac{1}{2}$$

2 Statistical Averages: Mean, Variance and Standard Deviation

$$E(X) \quad \text{Var}(X) \quad \sigma = \sqrt{\text{Var}(X)}$$

2.1 Mean Value (Expectation)

Let X be a random variable (r.v.) with pdf $f(x)$. Then its **mean value** is defined by

Definition — Mean / Expectation

$$E(X) = \bar{x} = \mu = \begin{cases} \sum x f(x), & \text{if } X \text{ is a DRV} \\ \int_{-\infty}^{\infty} x f(x) dx, & \text{if } X \text{ is a CRV} \end{cases}$$

This mean value is also known as the **expectation** or **expected value** of the r.v. X .

2.2 Variance and Standard Deviation

Definition — Variance

$$\text{Var}(X) = V(X) = E[(X - \bar{x})^2] = \begin{cases} \sum (x - \bar{x})^2 f(x), & \text{if } X \text{ is a DRV} \\ \int_{-\infty}^{\infty} (x - \bar{x})^2 f(x) dx, & \text{if } X \text{ is a CRV} \end{cases}$$

Standard Deviation

$$\text{S.D.} = \sigma = \sqrt{\text{Var}(X)} \quad \leftarrow \quad \text{measures how the values spread out}$$

2.3 Useful Results

Results

1. $E(aX) = a E(X)$, where a is a constant.
2. $\text{Var}(aX \pm b) = a^2 \text{Var}(X)$, where a and b are constants.
3. $\text{Var}(X) = E(X^2) - (E(X))^2$

2.4 Worked Problems

Problem 2.1 — Expected Number of Tosses

A coin is tossed until a tail appears. What is the expectation of the number of tosses?

Solution — Setting up the PDF

Let X be a r.v. that denotes the number of tosses required to get a tail. Then we have the following pdf:

$X = x$	1	2	3	4	...
$P(X = x) = f(x)$	$\frac{1}{2}$	$\frac{1}{2^2}$ (HT)	$\frac{1}{2^3}$ (HHT)	$\frac{1}{2^4}$	...

Solution — Computing $E(X)$

$$\begin{aligned}
 E(X) &= \sum_x x f(x) = 1\left(\frac{1}{2}\right) + 2\left(\frac{1}{2^2}\right) + 3\left(\frac{1}{2^3}\right) + \cdots \\
 &= \frac{1}{2}\left(1 + 2\left(\frac{1}{2}\right) + 3\left(\frac{1}{2}\right)^2 + 4\left(\frac{1}{2}\right)^3 + \cdots\right) \\
 &= \frac{1}{2}(1 + 2y + 3y^2 + \cdots), \quad \text{where } y = \frac{1}{2} \\
 &= \frac{1}{2}(1 - y)^{-2} = \frac{1}{2}\left(1 - \frac{1}{2}\right)^{-2} = \frac{1}{2}\left(\frac{1}{2}\right)^{-2} = \frac{1}{2} \cdot 2^2 = 2
 \end{aligned}$$

$$E(X) = 2$$

On average, we require **2 tosses** to get a tail.

Problem 2.2 — Cars per Household

A market researcher at a major automobile company classified households by car ownership. The relative frequencies of the households for each category of ownership are as follows.

Number of cars per household	0	1	2	3	4	5
Relative frequencies	0.10	0.30	0.40	0.12	0.06	0.02

- Show that the given table of relative frequencies represents the pdf of the r.v. X (the number of cars per household).
- Find the mean, variance and standard deviation of X .

Solution — (a) The table is a pdf, and (b) the mean

It is clear from the table that all the frequency values are ≥ 0 and also

$$\sum (\text{all frequency values}) = 0.10 + 0.30 + 0.40 + 0.12 + 0.06 + 0.02 = 1$$

$\therefore$ The given table is the **pdf of the r.v. X** .

$$\begin{aligned}
 \therefore \text{Mean} &= E(X) = \sum_x x f(x) \\
 &= 0(0.10) + 1(0.30) + 2(0.40) + 3(0.12) + 4(0.06) + 5(0.02)
 \end{aligned}$$

$$E(X) = 1.8 \Rightarrow \text{the average number of cars per household is 1.8}$$

Solution — (b) Variance and standard deviation

$$\begin{aligned}
 \text{Var}(X) &= E(X^2) - (E(X))^2 = \sum_x x^2 f(x) - (1.8)^2 \\
 &= [0^2(0.10) + 1^2(0.30) + 2^2(0.40) + 3^2(0.12) \\
 &\quad + 4^2(0.06) + 5^2(0.02)] - (1.8)^2 \\
 &= 4.44 - (1.8)^2 = 4.44 - 3.24
 \end{aligned}$$

$$\text{Var}(X) = 1.2 \quad \text{S.D.} = \sigma = \sqrt{1.2} \Rightarrow \sigma = 1.09$$

Problem 2.3 — Average Profit per SIM Card

A mobile shop is promoting a sale in which the purchaser of a SIM card can roll a die and then deduct a rupee from the retail price for each dot shown on the rolled die. It is equally likely that the die will come up any integer from 1 to 6. The shop owner pays Rs. 5.00 for each SIM card and then prices it Rs. 9.00. During this special promotion, what will be the shop's average profit per SIM card sold?

Solution — Setting up the PDF

Let X denote the selling price of a SIM card. Since the die shows 1, 2, 3, 4, 5, 6 (each with probability $\frac{1}{6}$), the price Rs. 9 is reduced by that many rupees:

die shows	(1	2	3	4	5	6)
$X = x$	8	7	6	5	4	3
$f(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Solution — Computing the average profit

$$E(X) = \sum_x x f(x) = 8\left(\frac{1}{6}\right) + 7\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right)$$

$$E(X) = 5.5$$

The shop owner bought each SIM card at Rs. 5, so

$$\text{Profit on selling each SIM card} = 5.5 - 5 = 0.50$$

$$\text{Average profit} = \text{Rs. } 0.50 \text{ (50 paise) per SIM card}$$

Problem 2.4 — Mean and Variance of a Continuous r.v.

Find the mean and variance of the r.v. X whose pdf is given by

$$f(x) = \begin{cases} \frac{x+2}{8}, & -1 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Solution — Mean

Since X is a CRV,

$$\begin{aligned} \text{Mean} = E(X) &= \int_{-\infty}^{\infty} x f(x) dx = \int_{-1}^1 x \left(\frac{x+2}{8} \right) dx \\ &= \frac{1}{8} \int_{-1}^1 (x^2 + 2x) dx = \frac{1}{8} \left[\left(\frac{x^3}{3} \right)_{-1}^1 + 2 \left(\frac{x^2}{2} \right)_{-1}^1 \right] \\ &= \frac{1}{8} \left[\left(\frac{1}{3} - \left(-\frac{1}{3} \right) \right) + (1 - 1) \right] = \frac{1}{8} \left[\frac{2}{3} \right] = \frac{1}{12} \end{aligned}$$

$$\text{Mean} = E(X) = \frac{1}{12}$$

Solution — Variance

$\text{Var}(X) = E(X^2) - (E(X))^2$. Now,

$$\begin{aligned} E(X^2) &= \int_{-1}^1 x^2 f(x) dx = \int_{-1}^1 x^2 \left(\frac{x+2}{8} \right) dx \\ &= \frac{1}{8} \int_{-1}^1 (x^3 + 2x^2) dx = \frac{1}{8} \left[\frac{x^4}{4} + \frac{2x^3}{3} \right]_{-1}^1 \\ &= \frac{1}{8} \left[\left(\frac{1}{4} - \frac{1}{4} \right) + \frac{2}{3} (1 - (-1)) \right] = \frac{1}{8} \left[\frac{4}{3} \right] = \frac{1}{6} \end{aligned}$$

$$\therefore \text{Var}(X) = E(X^2) - (E(X))^2 = \frac{1}{6} - \left(\frac{1}{12} \right)^2 = \frac{1}{6} - \frac{1}{144}$$

$$\text{Var}(X) = \frac{23}{144} \approx 0.159$$

3 Moment Generating Function (MGF)

Definition — Moment Generating Function

Moments can easily be generated by the MGF, which is defined by

$$M_X(t) = E(e^{tX}) = \begin{cases} \sum e^{tx} f(x), & \text{if } X \text{ is a DRV} \\ \int_{-\infty}^{\infty} e^{tx} f(x) dx, & \text{if } X \text{ is a CRV} \end{cases}$$

3.1 Why It Generates Moments

Expansion of $M_X(t)$

$$\begin{aligned} M_X(t) &= E(e^{tX}) \\ &= E\left(1 + \frac{tX}{1!} + \frac{(tX)^2}{2!} + \frac{(tX)^3}{3!} + \cdots + \frac{(tX)^r}{r!} + \cdots\right) \\ &= 1 + \frac{t}{1!}E(X) + \frac{t^2}{2!}E(X^2) + \frac{t^3}{3!}E(X^3) + \cdots + \frac{t^r}{r!}E(X^r) + \cdots \\ &= 1 + \frac{t}{1!}\mu'_1 + \frac{t^2}{2!}\mu'_2 + \cdots + \frac{t^r}{r!}\mu'_r + \cdots \quad (\because \mu'_r = E(X^r)) \end{aligned}$$

$\Rightarrow \mu'_r = r^{\text{th}}$ moment about the origin.

$$\mu'_r = \text{Coeff. of } \frac{t^r}{r!} \text{ in the expansion of } M_X(t)$$

Results

$$\mu'_r = \left\{ \frac{d^r}{dt^r} M_X(t) \right\}_{t=0} \quad \mu'_1 = \text{mean of } X \quad \text{Var}(X) = \mu'_2 - (\mu'_1)^2$$

3.2 Worked Problems

Problem 3.1 — MGF of a Geometric-Type Discrete r.v.

Find the MGF of the r.v. X whose pdf is given by

$$f(x) = \frac{1}{2^x}, \quad x = 1, 2, 3, \dots$$

and hence find the mean and variance of X .

Solution — The MGF

Given X is a DRV with pdf $f(x) = \frac{1}{2^x}$, $x = 1, 2, 3, \dots$

$$\begin{aligned}
 \therefore \text{MGF} = M_X(t) &= E(e^{tX}) = \sum_x e^{tx} f(x) = \sum_{x=1,2,3,\dots} e^{tx} \frac{1}{2^x} \\
 &= \sum_{x=1,2,3,\dots} \left(\frac{e^t}{2}\right)^x = \frac{e^t}{2} + \left(\frac{e^t}{2}\right)^2 + \left(\frac{e^t}{2}\right)^3 + \dots \\
 &= \frac{e^t}{2} \left(1 + \frac{e^t}{2} + \left(\frac{e^t}{2}\right)^2 + \dots\right) \\
 &= \frac{e^t}{2} (1 + y + y^2 + \dots), \quad \text{where } y = \frac{e^t}{2} \\
 &= \frac{e^t}{2} (1 - y)^{-1} = \frac{e^t}{2} \left(1 - \frac{e^t}{2}\right)^{-1} = \frac{e^t}{2} \left(\frac{2}{2 - e^t}\right)
 \end{aligned}$$

$$M_X(t) = \frac{e^t}{2 - e^t}$$

Solution — Deduction — Mean

$$\begin{aligned}
 \text{Mean} = \mu'_1 &= \left\{ \frac{d}{dt} M_X(t) \right\}_{t=0} = \left\{ \frac{d}{dt} \left(\frac{e^t}{2 - e^t} \right) \right\}_{t=0} \\
 &= \left\{ \frac{(2 - e^t)e^t - e^t(-e^t)}{(2 - e^t)^2} \right\}_{t=0} = \left\{ \frac{2e^t - e^{2t} + e^{2t}}{(2 - e^t)^2} \right\}_{t=0} \\
 &= \left\{ \frac{2e^t}{(2 - e^t)^2} \right\}_{t=0} = \frac{2}{(2 - 1)^2} = 2
 \end{aligned}$$

$$\text{Mean} = 2$$

Solution — Deduction — Variance

Variance = $\mu'_2 - (\mu'_1)^2$, and

$$\begin{aligned}
 \mu'_2 &= \left\{ \frac{d^2}{dt^2} M_X(t) \right\}_{t=0} = \left\{ \frac{d}{dt} \left(\frac{d}{dt} M_X(t) \right) \right\}_{t=0} \\
 &= \left\{ \frac{d}{dt} \left(\frac{2e^t}{(2 - e^t)^2} \right) \right\}_{t=0} = \left\{ \frac{(2 - e^t)^2 2e^t - 2e^t \cdot 2(2 - e^t)(-e^t)}{(2 - e^t)^4} \right\}_{t=0} \\
 &= \frac{2 + 4}{1} = 6
 \end{aligned}$$

$$\therefore \text{Variance} = \mu'_2 - (\mu'_1)^2 = 6 - (2)^2$$

$$\text{Variance} = 2$$

Problem 3.2 — MGF of an Exponential r.v.

Find the MGF, mean and variance of the CRV X with pdf

$$f(x) = \begin{cases} \frac{1}{5} e^{-x/5}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Solution — The MGF

$$\begin{aligned} M_X(t) &= E(e^{tX}) = \int_0^\infty e^{tx} f(x) dx = \int_0^\infty e^{tx} \frac{1}{5} e^{-x/5} dx \\ &= \frac{1}{5} \int_0^\infty e^{-x(\frac{1}{5}-t)} dx = \frac{1}{5} \left[\frac{e^{-(\frac{1}{5}-t)x}}{-\left(\frac{1}{5}-t\right)} \right]_0^\infty \\ &= \frac{1}{5} \left[\frac{1}{-\left(\frac{1}{5}-t\right)} (0-1) \right] = \frac{1}{5} \cdot \frac{1}{\left(\frac{1-5t}{5}\right)} = \frac{1}{1-5t} \end{aligned}$$

$$M_X(t) = \frac{1}{1-5t}$$

Solution — Deductions — Mean and Variance

$$\begin{aligned} \text{Mean} = \mu'_1 &= \left\{ \frac{d}{dt} M_X(t) \right\}_{t=0} = \left\{ \frac{d}{dt} \left(\frac{1}{1-5t} \right) \right\}_{t=0} \\ &= \left\{ (-1)(1-5t)^{-2} (-5) \right\}_{t=0} = \left\{ \frac{5}{(1-5t)^2} \right\}_{t=0} = 5 \end{aligned}$$

$$\text{Mean} = \mu'_1 = 5$$

$$\begin{aligned} \mu'_2 &= \left\{ \frac{d^2}{dt^2} M_X(t) \right\}_{t=0} = \left\{ \frac{d}{dt} \left(\frac{5}{(1-5t)^2} \right) \right\}_{t=0} = 50 \\ \Rightarrow \text{Variance} &= \mu'_2 - (\mu'_1)^2 = 50 - (5)^2 = 25 \end{aligned}$$

$$\text{Variance} = 25$$

4 Some Special Distributions

Relative to some probability function:

Discrete

- Binomial distribution
- Poisson distribution

Continuous

- Normal (or) Gaussian distribution
- Gamma (or) Erlang distribution

The Bernoulli Variate

- Takes only two values $X = 0$ (failure) and $X = 1$ (success)
- $P(X = 1) = p$ and $P(X = 0) = q$
- PDF $f(x) = P(X = x) = p^x q^{1-x}$, $x = 0, 1$
- Repeated n times, the PDF is $f(x) = P(X = x) = \binom{n}{x} p^x q^{n-x}$, $x = 0, 1, 2, \dots, n$

PART I

BINOMIAL DISTRIBUTION

5 Binomial Distribution

5.1 Theory

Bernoulli random variable — step by step

- Suppose X is a discrete random variable (DRV) taking only two values

$$X = 0 \text{ (failure), } X = 1 \text{ (success)}$$

$X = x$	0	1
$P(X = x)$	q	p

- such that

$$P(X = 1) = p, \quad P(X = 0) = q, \quad p + q = 1$$

- and its p.d.f. is

$$P(X = x) = p^x q^{1-x}, \quad x = 0, 1$$

From Bernoulli trials to the binomial p.d.f.

- The random variable which takes only the two values 0 and 1 is called a **Bernoulli random variable**.
- If we repeat a Bernoulli trial ' n ' times, then the probability of getting ' x ' successes in n trials is given by the p.d.f. ${}^nC_x p^x q^{n-x}$.

Definition — Binomial Distribution

A discrete random variable X is said to follow the **Binomial distribution** if its p.d.f. is of the form

$$f(x) = P(X = x) = \begin{cases} {}^nC_x p^x q^{n-x}, & x = 0, 1, 2, \dots, n \\ 0, & \text{otherwise} \end{cases}$$

Notation

$$X \sim B(n, p)$$

where n and p are the **parameters** of the Binomial distribution.

Note

Putting $x = 0, 1, 2, \dots, n$ in the p.d.f. gives

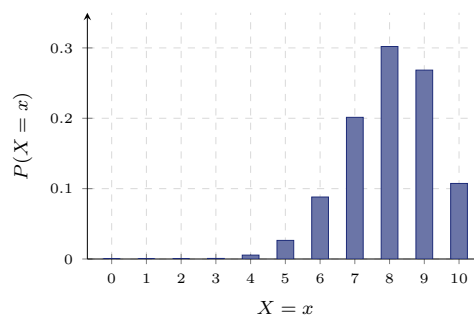
$$q^n, \quad {}^nC_1 p q^{n-1}, \quad {}^nC_2 p^2 q^{n-2}, \quad \dots, \quad p^n$$

which are the successive terms of the binomial expansion $(q+p)^n$ — hence the name **Binomial distribution**.

Shape of the binomial distribution

Example: $n = 10, p = 0.8$

- $P(X = x) = {}^{10}C_x (0.8)^x (0.2)^{10-x}$
- Total mass $= (q + p)^{10} = 1$
- Peak (mode) near $np = 8$
- Skewed to the left because $p > \frac{1}{2}$



Binomial distribution with $n = 10, p = 0.8$

5.2 Standard Results

Results for the Binomial Distribution

MGF: $M_X(t) = (q + pe^t)^n$

Mean: $\mu = np$

Variance: $\sigma^2 = npq$

Standard deviation: $\sigma = \sqrt{npq}$

Each of these is derived line by line below.

Result 1 — MGF of the binomial distribution is $(q + pe^t)^n$

Proof

Given $X \sim B(n, p)$, the p.d.f. is $f(x) = P(X = x) = {}^nC_x p^x q^{n-x}$, $x = 0, 1, 2, \dots, n$. Then

$$\begin{aligned}
 \text{MGF} &= M_X(t) = E(e^{tX}) = \sum_{x=0}^n e^{tx} f(x) = \sum_{x=0}^n e^{tx} {}^nC_x p^x q^{n-x} \\
 &= \sum_{x=0}^n {}^nC_x (pe^t)^x q^{n-x} \\
 &= q^n + {}^nC_1 (pe^t) q^{n-1} + {}^nC_2 (pe^t)^2 q^{n-2} + \dots + (pe^t)^n
 \end{aligned}$$

$$M_X(t) = (q + pe^t)^n$$

Reason used

$$(x + a)^n = x^n + {}^nC_1 x^{n-1} a + {}^nC_2 x^{n-2} a^2 + \dots + a^n$$

Deduction 1 — Mean of the binomial distribution

$$\begin{aligned}
 \text{Mean} &= \mu'_1 = \left\{ \frac{d}{dt} M_X(t) \right\}_{t=0} = \left\{ \frac{d}{dt} (q + pe^t)^n \right\}_{t=0} \\
 &= \left\{ n (q + pe^t)^{n-1} pe^t \right\}_{t=0} = n (q + p)^{n-1} p
 \end{aligned}$$

$$\text{Mean} = np$$

since $p + q = 1$

Deduction 2 — Variance of the binomial distribution

$$\text{Variance} = \mu'_2 - (\mu'_1)^2$$

$$\begin{aligned}\mu'_2 &= \left\{ \frac{d^2}{dt^2} M_X(t) \right\}_{t=0} = \left\{ \frac{d}{dt} \left(\frac{d}{dt} M_X(t) \right) \right\}_{t=0} \\ &= \left\{ \frac{d}{dt} \left(\underbrace{np e^t}_u \underbrace{(q + pe^t)^{n-1}}_v \right) \right\}_{t=0} \\ &= \left(np e^t (n-1) (q + pe^t)^{n-2} pe^t + (q + pe^t)^{n-1} np e^t \right)_{t=0}\end{aligned}$$

Putting $t = 0$ and using $q + p = 1$:

$$\mu'_2 = np(n-1)p + np = n^2p^2 - np^2 + np$$

$$\text{Variance} = \mu'_2 - (\mu'_1)^2 = n^2p^2 - np^2 + np - (np)^2 = -np^2 + np = np(1-p)$$

$$\text{Variance} = npq, \quad \text{S.D.} = \sqrt{npq}$$

since $p + q = 1 \Rightarrow 1 - p = q$

Summary Card — Binomial

Quantity	Binomial $X \sim B(n, p)$
p.d.f.	${}^nC_x p^x q^{n-x}, x = 0, 1, \dots, n$
MGF	$(q + pe^t)^n$
Mean	np
Variance	npq
S.D.	$\sqrt{npq}$
Condition	$p + q = 1$

5.3 Worked Problems

Problem 5.1 — Families with Four Children

Out of **800 families** with **4 children** each, how many families would be expected to have

- (i) 2 boys and 2 girls,
- (ii) at least 1 boy,
- (iii) at most 2 girls,
- (iv) children of both sexes?

Assume equal probabilities for boys and girls.

Solution — Setting up

Let X be the number of boys in a family.

$$P(\text{getting a boy}) = \frac{1}{2}; \quad P(\text{getting a girl}) = \frac{1}{2}$$

$$\Rightarrow p = \frac{1}{2}; \quad q = \frac{1}{2}$$

A family has 4 children $\Rightarrow n = 4$, so $X \sim B\left(4, \frac{1}{2}\right)$ and

$$P(X = x) = {}^4C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{4-x} = {}^4C_x \left(\frac{1}{2}\right)^4, \quad x = 0, 1, 2, 3, 4$$

$$P(X = x) = \frac{1}{16} {}^4C_x, \quad x = 0, 1, 2, 3, 4$$

x	0	1	2	3	4
4C_x	1	4	6	4	1
$P(X = x)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

Solution — (i) 2 boys and 2 girls

$$P(2 \text{ boys and 2 girls}) = P(X = 2) = \frac{1}{16} {}^4C_2 = \frac{6}{16} = \frac{3}{8}$$

$${}^4C_2 = \frac{4!}{2!2!} = 6$$

$\therefore$ Out of 800 families, the number having 2 boys and 2 girls

$$= \frac{3}{8} \times 800 = 300 \text{ families}$$

Solution — (ii) At least one boy

$$P(\text{at least one boy}) = P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{1}{16} {}^4C_0 = \frac{15}{16}$$

$\therefore$ The number of families having at least one boy

$$= \frac{15}{16} \times 800 = 750 \text{ families}$$

Solution — (iii) At most 2 girls

Let Y denote the number of girls, so $Y = 4 - X$.

$$\begin{aligned} P(\text{at most 2 girls}) &= P(Y \leq 2) = P(Y = 0) + P(Y = 1) + P(Y = 2) \\ &= P(X = 4) + P(X = 3) + P(X = 2) \\ &= \frac{1}{16} {}^4C_4 + \frac{1}{16} {}^4C_3 + \frac{1}{16} {}^4C_2 = \frac{1 + 4 + 6}{16} = \frac{11}{16} \end{aligned}$$

$\therefore$ The number of families having at most 2 girls

$$= \frac{11}{16} \times 800 = 550 \text{ families}$$

Solution — (iv) Children of both sexes

$$\begin{aligned} P(\text{children of both sexes}) &= P(X = 1) + P(X = 2) + P(X = 3) \\ &= \frac{1}{16} {}^4C_1 + \frac{1}{16} {}^4C_2 + \frac{1}{16} {}^4C_3 = \frac{4 + 6 + 4}{16} = \frac{14}{16} = \frac{7}{8} \end{aligned}$$

$\therefore$ The number of families having children of both sexes

$$= \frac{7}{8} \times 800 = 700 \text{ families}$$

Problem 5.2 — Noisy Transmission Channel

A noisy transmission channel has a per digit error probability $p = 0.01$. Calculate the probability that **more than 1 error** occurs in **10 received digits**.

Solution — Setting up

Given $p = 0.01$, $n = 10$, so $q = 1 - p = 0.99$. Let X be the number of errors in the received digits; then $X \sim B(n, p)$ and

$$P(X = x) = {}^{10}C_x (0.01)^x (0.99)^{10-x}, \quad x = 0, 1, 2, \dots, 10.$$

Solution — Computation

$\therefore$ $P(\text{more than 1 error in 10 received digits})$

$$\begin{aligned} &= P(X > 1) = 1 - P(X \leq 1) = 1 - (P(X = 0) + P(X = 1)) \\ &= 1 - \left({}^{10}C_0 (0.01)^0 (0.99)^{10} + {}^{10}C_1 (0.01) (0.99)^9 \right) \\ &= 1 - \left((0.99)^{10} + 10 (0.01) (0.99)^9 \right) = 1 - (0.9044 + 0.0914) = 1 - 0.9957 \end{aligned}$$

$$P(X > 1) = 0.0043$$

Problem 5.3 — Finding the Parameter p

In a binomial distribution consisting of **5 independent trials**, the probabilities of **1** and **2** successes are 0.4096 and 0.2048 respectively. Find the parameter p of the distribution.

Solution — Setting up

Given $n = 5$ and $X \sim B(n, p)$, so $P(X = x) = {}^5C_x p^x q^{5-x}$, $x = 0, 1, 2, 3, 4, 5$. Also given

$$P(X = 1) = 0.4096 \quad \text{and} \quad P(X = 2) = 0.2048,$$

that is,

$${}^5C_1 p q^4 = 0.4096 \quad \text{and} \quad {}^5C_2 p^2 q^3 = 0.2048.$$

Solution — Solving for p

$$5 p (1 - p)^4 = 0.4096 \quad \text{and} \quad 10 p^2 (1 - p)^3 = 0.2048$$

Dividing the first equation by the second:

$$\begin{aligned} \frac{5 p (1 - p)^4}{10 p^2 (1 - p)^3} &= \frac{0.4096}{0.2048} \quad \Rightarrow \quad \frac{1 - p}{2p} = 2 \\ \Rightarrow \quad 1 - p &= 4p \quad \Rightarrow \quad 5p = 1 \end{aligned}$$

$$p = \frac{1}{5}$$

$${}^5C_1 = 5, \quad {}^5C_2 = \frac{5!}{2!3!} = 10$$

Problem 5.4 — Mean and Variance from the MGF

If the MGF of a random variable X is $(0.4e^t + 0.6)^8$, find its **mean** and **variance**.

Solution — Identifying the distribution

Given $M_X(t) = (0.4e^t + 0.6)^8$, which is of the form $(q + pe^t)^n$ — the MGF of a Binomial distribution. Hence

$$p = 0.4, \quad q = 0.6 \quad \text{and} \quad n = 8.$$

Using the standard results,

$$\text{Mean} = np = 8 \times 0.4 = 3.2$$

$$\text{Variance} = npq = 8 \times 0.4 \times 0.6 = 1.92$$

$$\text{Mean} = 3.2, \quad \text{Variance} = 1.92$$

Problem 5.5 — Basketball Free Throws

Over a long period of time, it has been observed that a professional basketball player can make a free throw on a given trial with probability **0.8**. Suppose he shoots **four free throws**.

- (a) What is the probability that he will make **exactly two** free throws?
- (b) What is the probability that he will make **at least one** free throw?

Solution — Setting up

Let X be the number of free throws made out of the four attempts. Given $p = 0.8$, $n = 4$, so $q = 1 - p = 0.2$. Each throw is an independent Bernoulli trial, hence $X \sim B(4, 0.8)$ and

$$P(X = x) = {}^4C_x(0.8)^x(0.2)^{4-x}, \quad x = 0, 1, 2, 3, 4.$$

Solution — (a) Exactly two free throws

$$P(X = 2) = {}^4C_2(0.8)^2(0.2)^2 = 6 \times (0.64) \times (0.04)$$

$$P(X = 2) = 0.1536$$

$${}^4C_2 = \frac{4!}{2!2!} = 6$$

Solution — (b) At least one free throw

$$P(X \geq 1) = 1 - P(X = 0) = 1 - {}^4C_0(0.8)^0(0.2)^4 = 1 - (0.2)^4 = 1 - 0.0016$$

$$P(X \geq 1) = 0.9984$$

$${}^4C_0 = 1 \quad \text{and} \quad (0.8)^0 = 1$$

PART II

POISSON DISTRIBUTION

6 Poisson Distribution

6.1 Theory

Poisson as the Limiting Case of the Binomial

$$P(X = x) = \lim_{\substack{n \rightarrow \infty \\ p \rightarrow 0 \\ (np = \lambda)}} {}^nC_x p^x q^{n-x} = \boxed{\frac{e^{-\lambda} \lambda^x}{x!}} \quad \text{where } \lambda = np$$

- Limiting case of the Binomial distribution
- $n \rightarrow \infty$, $p \rightarrow 0$, $np = \lambda$ a constant
- Apply when n is **large** and p is **small**

In the expansion one uses $n^x \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \rightarrow n^x$ as $n \rightarrow \infty$.

Definition — Poisson Distribution

A discrete random variable X is said to follow the **Poisson distribution** if its p.d.f. is of the form

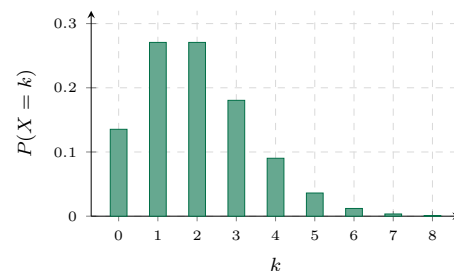
$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

Notation

$$X \sim P(\lambda)$$

where λ is the **parameter**; also

$$\boxed{\lambda = np}$$



Poisson distribution with $\lambda = 2$

6.2 Standard Results

Result — MGF of the Poisson distribution is $e^{\lambda(e^t - 1)}$

Proof

$$\begin{aligned}
 \text{MGF} &= M_X(t) = E(e^{tX}) = \sum_x e^{tx} f(x) = \sum_{x=0}^{\infty} e^{tx} \frac{e^{-\lambda} \lambda^x}{x!} \\
 &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!} \\
 &= e^{-\lambda} \left[1 + \frac{\lambda e^t}{1!} + \frac{(\lambda e^t)^2}{2!} + \frac{(\lambda e^t)^3}{3!} + \dots \right] = e^{-\lambda} e^{\lambda e^t}
 \end{aligned}$$

$$\text{MGF} = M_X(t) = e^{\lambda(e^t - 1)}$$

$$\text{using } e^y = 1 + \frac{y}{1!} + \frac{y^2}{2!} + \frac{y^3}{3!} + \dots$$

Deduction 1 — Mean of the Poisson distribution

$$\text{Mean} = \left\{ \frac{d}{dt} M_X(t) \right\}_{t=0} = \mu'_1 = \left\{ \frac{d}{dt} (e^{\lambda(e^t - 1)}) \right\}_{t=0} = \left\{ e^{\lambda(e^t - 1)} \cdot \lambda e^t \right\}_{t=0}$$

$$\text{Mean} = \lambda = \mu'_1$$

$$\text{at } t = 0: e^t = 1 \Rightarrow e^{\lambda(1-1)} = e^0 = 1$$

Deduction 2 — Variance of the Poisson distribution

$$\text{Variance} = \mu'_2 - (\mu'_1)^2$$

$$\begin{aligned}
 \mu'_2 &= \left\{ \frac{d^2}{dt^2} M_X(t) \right\}_{t=0} = \left\{ \frac{d}{dt} \left(\frac{d}{dt} M_X(t) \right) \right\}_{t=0} \\
 &= \left\{ \frac{d}{dt} \left(\underbrace{\lambda e^t}_u \underbrace{e^{\lambda(e^t - 1)}}_v \right) \right\}_{t=0} = \lambda(\lambda + 1) = \lambda^2 + \lambda
 \end{aligned}$$

$$\text{Variance} = \mu'_2 - (\mu'_1)^2 = \lambda^2 + \lambda - \lambda^2$$

$$\text{Variance} = \lambda$$

Characteristic property

$$\text{Mean} = \text{Variance} = \lambda$$

Summary Card — Poisson

Quantity	Poisson $X \sim P(\lambda)$
p.d.f.	$\frac{e^{-\lambda}\lambda^x}{x!}, x = 0, 1, 2, \dots$
MGF	$e^{\lambda(e^t - 1)}$
Mean	λ
Variance	λ
S.D.	$\sqrt{\lambda}$
Link to Binomial	$\lambda = np$ (n large, p small)

6.3 Worked Problems

Problem 6.1 — Defective Pins

A manufacturer of pins knows that **2%** of his products are defective. If he sells pins in **boxes of 100** and guarantees that **not more than 4 pins** will be defective, what is the probability that a box will meet the guaranteed quality?

Solution — Setting up

Given $p = 0.02$ and $n = 100$; since n is large and p is small we may use the Poisson p.d.f. with

$$\lambda = np = (0.02) \times 100 = 2.$$

$$\lambda = 2$$

Let X be the number of defectives in a box, $X \sim P(\lambda)$; then

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!} = \frac{e^{-2}2^x}{x!}, \quad x = 0, 1, 2, 3, \dots$$

Solution — Computation

$\therefore P(\text{a box will fail to meet its guaranteed quality})$

$$\begin{aligned}
 &= P(X > 4) = 1 - P(X \leq 4) \\
 &= 1 - [P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)] \\
 &= 1 - \left[\frac{e^{-2}2^0}{0!} + \frac{e^{-2}2^1}{1!} + \frac{e^{-2}2^2}{2!} + \frac{e^{-2}2^3}{3!} + \frac{e^{-2}2^4}{4!} \right] \\
 &= 1 - e^{-2} \left[1 + 2 + 2 + \frac{4}{3} + \frac{2}{3} \right] = 1 - e^{-2} [7]
 \end{aligned}$$

$$P(X > 4) = 0.0526$$

$\therefore$ The probability that a box *will* meet the guaranteed quality

$$= 1 - 0.0526 = 0.9474$$

Problem 6.2 — Car Hire Firm

A car hire firm has **two cars** which it hires out day by day. The number of demands for a car on each day is distributed as a **Poisson variate with mean 1.5**. Calculate the proportion of days on which

- (i) neither car is used,
- (ii) some demand is refused.

Solution — Setting up

Let X be the number of demands for a car on each day; it follows $P(\lambda)$ with mean $\lambda = 1.5$, so $X \sim P(1.5)$ and

$$P(X = x) = \frac{e^{-1.5}(1.5)^x}{x!}, \quad x = 0, 1, 2, \dots$$

But the firm has only two cars.

Solution — (i) Neither car is used

$$P(\text{neither car is used}) = P(X = 0) = \frac{e^{-1.5}(1.5)^0}{0!} = e^{-1.5}$$

$$P(X = 0) = 0.223$$

Solution — (ii) Some demand is refused

$$\begin{aligned} P(\text{some demand is refused}) &= P(X > 2) = 1 - P(X \leq 2) \\ &= 1 - [P(X = 0) + P(X = 1) + P(X = 2)] \\ &= 1 - \left[\frac{e^{-1.5}(1.5)^0}{0!} + \frac{e^{-1.5}(1.5)^1}{1!} + \frac{e^{-1.5}(1.5)^2}{2!} \right] \\ &= 1 - e^{-1.5} \left[1 + 1.5 + \frac{(1.5)^2}{2} \right] \end{aligned}$$

$$P(X > 2) = 0.1911$$

Problem 6.3 — Typographical Errors in a Book

In a book of **520 pages**, **390 typographical errors** occur. Assuming the Poisson law for the number of errors per page, find the probability that a **random sample of 5 pages** will contain **no error**.

Solution — Setting up

Let X be the number of errors per page; it follows $P(\lambda)$, where $\lambda = \text{mean} = \text{average error per page}$:

$$\lambda = \frac{390}{520} = \frac{3}{4} = 0.75 \quad \implies \quad X \sim P(0.75)$$

so that

$$P(X = x) = \frac{e^{-0.75}(0.75)^x}{x!}, \quad x = 0, 1, 2, 3, \dots$$

Recall $\lambda = np$ and $\lambda \rightarrow \text{mean}$.

Solution — Computation

$\therefore P(\text{a random sample of 5 pages will contain no error})$

$$\begin{aligned} &= P((X = 0) \cap (X = 0) \cap (X = 0) \cap (X = 0) \cap (X = 0)) \\ &= \underbrace{P(X = 0) \times P(X = 0) \times \dots \times P(X = 0)}_{5 \text{ times}} = (P(X = 0))^5 \\ &= \left(\frac{e^{-0.75}(0.75)^0}{0!} \right)^5 = (e^{-0.75})^5 \\ &= e^{-3.75} = 0.0235 \end{aligned}$$

Problem 6.4 — Traffic Accidents

The mean number of traffic accidents occurring on a specific stretch of highway is **two per week**. Assume the occurrence of accidents follows a Poisson distribution with parameter $\lambda = 2$.

- Calculate the probability of **no accidents** happening on this highway segment during a **1-week** period.
- Determine the probability of having **at most three accidents** on this highway segment during a **2-week** period.

Solution — Setting up

Let X be the number of accidents in a **1-week** period, $X \sim P(\lambda)$ with $\lambda = 2$:

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!} = \frac{e^{-2}2^x}{x!}, \quad x = 0, 1, 2, \dots$$

Key idea for part (b)

For a Poisson process the mean is **proportional to the length of the interval**. Over 2 weeks the parameter therefore becomes $\lambda' = 2 \times 2 = 4$.

Solution — (a) No accidents in a 1-week period

$$P(\text{no accidents in 1 week}) = P(X = 0) = \frac{e^{-2}2^0}{0!} = e^{-2}$$

$$P(X = 0) = 0.1353$$

since $2^0 = 1$ and $0! = 1$

Solution — (b) At most three accidents in 2 weeks

Let Y be the number of accidents in a 2-week period, so $Y \sim P(4)$.

$$\begin{aligned} P(Y \leq 3) &= P(Y = 0) + P(Y = 1) + P(Y = 2) + P(Y = 3) \\ &= e^{-4} \left[\frac{4^0}{0!} + \frac{4^1}{1!} + \frac{4^2}{2!} + \frac{4^3}{3!} \right] = e^{-4} \left[1 + 4 + 8 + \frac{32}{3} \right] = e^{-4} (23.6667) \end{aligned}$$

$$P(Y \leq 3) = 0.4335$$

6.4 Binomial vs. Poisson — At a Glance**Consolidated Comparison**

	Binomial $B(n, p)$	Poisson $P(\lambda)$
p.d.f.	${}^nC_x p^x q^{n-x}$	$\frac{e^{-\lambda} \lambda^x}{x!}$
Range of x	$0, 1, 2, \dots, n$ (finite)	$0, 1, 2, \dots$ (infinite)
Parameters	n, p	λ
MGF	$(q + pe^t)^n$	$e^{\lambda(e^t - 1)}$
Mean	np	λ
Variance	npq	λ
S.D.	$\sqrt{npq}$	$\sqrt{\lambda}$
Mean vs. Variance	Mean > Variance	Mean = Variance
Relation	$B(n, p) \rightarrow P(\lambda)$ as $n \rightarrow \infty, p \rightarrow 0, np = \lambda$	

PART III

NORMAL (GAUSSIAN) DISTRIBUTION

7 Normal Distribution

7.1 Theory

Binomial $\longrightarrow$ Normal distribution

In $B(n, p)$, when **neither p nor q is small**, i.e. np is large, Stirling's formula $n! = n^n e^{-n} \sqrt{2\pi n}$ gives

$$f(x) = \binom{n}{x} p^x q^{n-x} \approx \frac{1}{\sqrt{2\pi npq}} e^{-\frac{(x-np)^2}{2npq}}.$$

Since $\mu = np$ and $\sigma^2 = npq$,

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad P(x_1 \leq X \leq x_2) = \int_{x_1}^{x_2} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

Definition — Normal (Gaussian) Distribution

A CRV X follows the **normal (Gaussian)** distribution if its PDF is

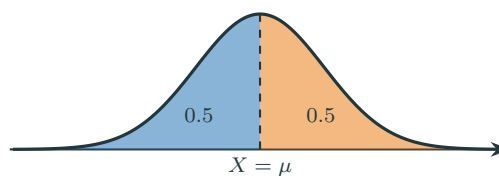
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$$

and we write $X \sim N(\mu, \sigma^2)$.

Results

(i) MGF = $e^{t\mu + \frac{\sigma^2 t^2}{2}}$ (ii) Mean = μ and Variance = σ^2

Properties of the normal curve



1. Symmetric about $X = \mu$ and bell shaped.
2. Total area under the curve is **unity**.
3. Area on either side of $X = \mu$ is **0.5**.

Standard normal variate

Let $X \sim N(\mu, \sigma^2)$ and put

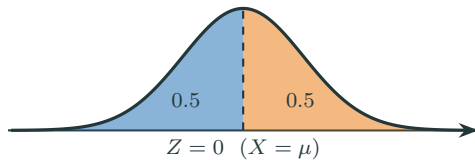
$$Z = \frac{X - \mu}{\sigma} \quad \implies \quad E(Z) = 0, \quad \text{Var}(Z) = 1$$

so that $Z \sim N(0, 1)$ is the **standard normal variate**, with PDF

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty.$$

The curve is symmetric about $Z = 0$ and bell shaped.

Areas under the standard normal curve



$$P(-\infty < Z < 0) = 0.5$$

$$P(0 < Z < \infty) = 0.5$$

$$P(-\infty < Z < \infty) = 1$$

By symmetry,

$$P(-a < Z < 0) = P(0 < Z < a)$$

Normal tables give $P(0 < Z < a)$.

7.2 Worked Problems

Problem 7.1 — Heights of Soldiers

Assume that the mean height of soldiers is 68.22 inches with variance 10.88 inches. How many soldiers in a regiment of 2000 soldiers would you expect to be over 6 feet tall? Assume heights are normally distributed.

Solution — Step 1 — Data

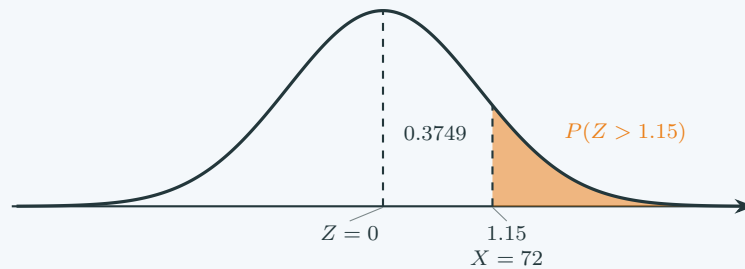
Let $X \rightarrow$ height of soldiers, $X \sim N(\mu, \sigma^2)$

$$\mu = 68.22 \text{ inches}, \quad \sigma^2 = 10.88, \quad \sigma = \sqrt{10.88} = 3.2985$$

Since 6 feet = 72 inches, we need $P(\text{height more than 6 feet}) = P(X > 72)$.

Solution — Step 2 — Standardise

$$\begin{aligned} P(X > 72) &= P\left(\frac{X - \mu}{\sigma} > \frac{72 - \mu}{\sigma}\right) = P\left(Z > \frac{72 - 68.22}{\sqrt{10.88}}\right) = P\left(Z > \frac{3.78}{3.2985}\right) \\ &= P(Z > 1.146) \approx P(Z > 1.15) \end{aligned}$$

Solution — Step 3 — The normal curve

$$P(Z > 1.15) = 0.5 - P(0 < Z < 1.15) = 0.5 - 0.3749 = \mathbf{0.1251}$$

Solution — Step 4 — Answer

$$P(X > 6 \text{ feet}) = 0.1251$$

The number of soldiers, in a regiment of 2000, having height more than 6 feet

$$= 2000 \times 0.1251 = 250.2$$

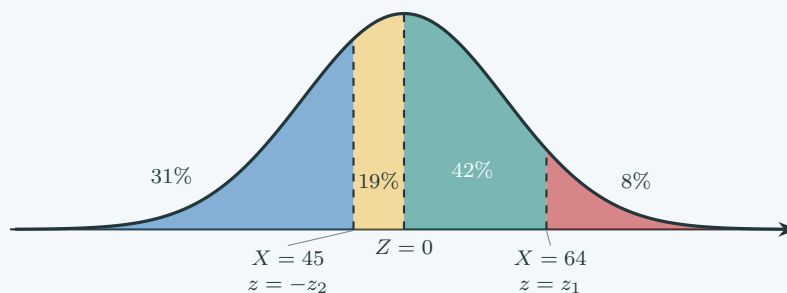
$$\approx 250 \text{ soldiers}$$

Problem 7.2 — Mean and S.D. from Percentages

In a normal distribution 31% of the items are under 45 and 8% of the items are over 64. Find the mean and standard deviation of the distribution.

Solution — Step 1 — The picture

Given $X \sim N(\mu, \sigma^2)$ with $P(X < 45) = 0.31$ and $P(X > 64) = 0.08$. Since $31\% < 50\%$, the value 45 lies to the **left** of μ ; since $8\% < 50\%$, the value 64 lies to the **right** of μ .

**Solution — Step 2 — Read the z values**

Let $Z = z_1$ and $Z = -z_2$ be the points corresponding to $X = 64$ and $X = 45$. From the figure:

$$P(0 < Z < z_1) = 42\% = 0.42 \implies \boxed{z_1 = 1.41}$$

$$P(-z_2 < Z < 0) = 19\% \implies P(0 < Z < z_2) = 0.19 \text{ (symmetry)} \implies \boxed{z_2 = 0.50}$$

(both from the normal table)

Solution — Step 3 — Two equations

Using $Z = \frac{X - \mu}{\sigma}$:

$$z_1 = \frac{64 - \mu}{\sigma} \quad 1.41 = \frac{64 - \mu}{\sigma}$$

$$\boxed{\mu + 1.41\sigma = 64} \quad (1)$$

$$-z_2 = \frac{45 - \mu}{\sigma} \quad -0.50 = \frac{45 - \mu}{\sigma}$$

$$\boxed{\mu - 0.50\sigma = 45} \quad (2)$$

Solution — Step 4 — Solve

(1) – (2):

$$1.91\sigma = 19 \implies \sigma = 9.9476$$

Substituting in (2):

$$\mu = 45 + 0.50(9.9476) = 45 + 4.974 = 49.97$$

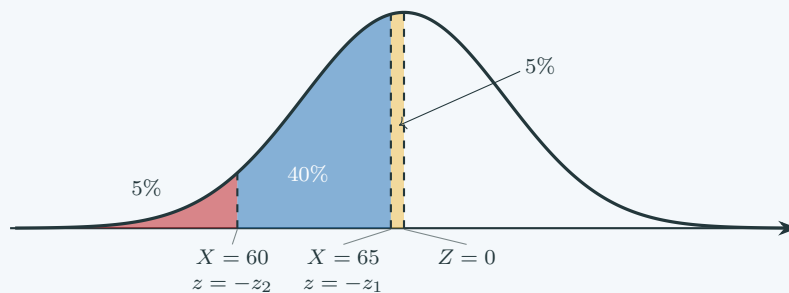
$$\text{Mean} = \mu \approx 50, \quad \text{S.D.} = \sigma \approx 10$$

Problem 7.3 — Heights of a Group of Men

Of a large group of men, 5% are under 60 inches in height and 40% are between 60 and 65 inches. Assuming a normal distribution, find the mean height and the standard deviation.

Solution — Step 1 — The picture

Given $P(X < 60) = 0.05$ and $P(60 < X < 65) = 0.40$, so $P(X < 65) = 0.45 < 0.5$. Hence **both** 60 and 65 lie to the left of μ .

**Solution — Step 2 — Read the z values**

Let $Z = -z_1$ and $Z = -z_2$ be the values corresponding to $X = 65$ and $X = 60$. From the figure:

$$\begin{aligned} P(-z_1 < Z < 0) &= 5\% \\ \Rightarrow P(0 < Z < z_1) &= 0.05 \\ \Rightarrow \boxed{z_1 = 0.13} \end{aligned}$$

$$\begin{aligned} P(-z_2 < Z < 0) &= 45\% \\ \Rightarrow P(0 < Z < z_2) &= 0.45 \\ \Rightarrow \boxed{z_2 = 1.65} \end{aligned}$$

Solution — Step 3 — Two equations

By the relation $Z = \frac{X - \mu}{\sigma}$:

$$-z_1 = \frac{65 - \mu}{\sigma} \quad -0.13 = \frac{65 - \mu}{\sigma}$$

$$\boxed{\mu - 0.13\sigma = 65} \quad (1)$$

$$-z_2 = \frac{60 - \mu}{\sigma} \quad -1.65 = \frac{60 - \mu}{\sigma}$$

$$\boxed{\mu - 1.65\sigma = 60} \quad (2)$$

Solution — Step 4 — Solve

(1) – (2):

$$1.52\sigma = 5 \implies \sigma = 3.289$$

Substituting in (1):

$$\mu = 65 + 0.13(3.289) = 65 + 0.4276 = 65.43$$

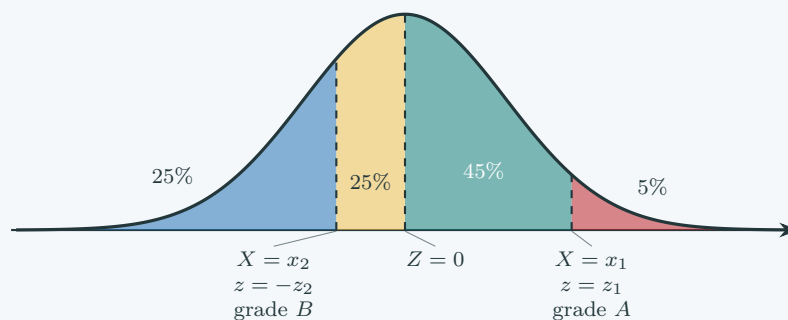
$$\text{Mean height} = 65.42'', \quad \text{S.D.} = 3.29''$$

Problem 7.4 — Examination Grades

A set of examination marks is approximately normally distributed with mean 75 and S.D. 5. If the top 5% of students get grade A and the bottom 25% get grade B, what mark is the lowest of A and what mark is the highest of B?

Solution — Step 1 — The picture

$X \rightarrow$ examination marks, $X \sim N(\mu, \sigma^2)$ with $\mu = 75$, $\sigma = 5$.

**Solution — Step 2 — Read the z values**

Let $X = x_1$ and $X = x_2$ be the lowest mark of grade A and the highest mark of grade B; let $Z = z_1$ and $Z = -z_2$ be the corresponding points.

$$P(0 < Z < z_1) = 45\% = 0.45 \implies \boxed{z_1 = 1.65}$$

$$P(-z_2 < Z < 0) = 25\% \implies P(0 < Z < z_2) = 0.25 \implies \boxed{z_2 = 0.68}$$

(both from the normal table)

Solution — Step 3 — Back-transform

Using $Z = \frac{X - \mu}{\sigma}$:

$$z_1 = \frac{x_1 - \mu}{\sigma} \quad 1.65 = \frac{x_1 - 75}{5}$$

$$x_1 = 75 + 8.25 = 83.25$$

$$-z_2 = \frac{x_2 - \mu}{\sigma} \quad -0.68 = \frac{x_2 - 75}{5}$$

$$x_2 = 75 - 3.40 = 71.60$$

Lowest mark of grade A ≈ 83 , Highest mark of grade B ≈ 72

Problem 7.5 — Examination Divisions

In an examination a student passes if he secures 30% or more marks. He is placed in the first, second and third divisions if he secures between 60% and 80%, between 45% and 60%, and between 30% and 45% respectively. He gets a distinction if he secures 80% or more. It is noticed that 10% of the students failed and 5% obtained a distinction. Calculate the percentage of students placed in the **second division**.

Solution — Step 1 — Data

Let $X \rightarrow$ marks obtained, $X \sim N(\mu, \sigma^2)$.

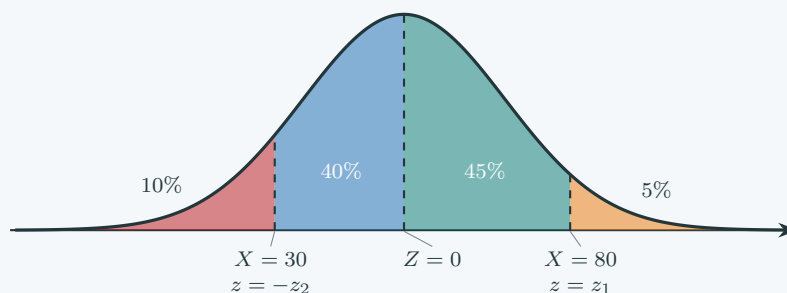
$$\text{Failed} \Rightarrow \text{marks below 30} : \quad P(X < 30) = 10\% = 0.10$$

$$\text{Distinction} \Rightarrow \text{marks above 80} : \quad P(X > 80) = 5\% = 0.05$$

Second division $\Rightarrow 45 < X < 60$ (this is what we must find).

Solution — Step 2 — The picture

Since $10\% < 50\%$, $X = 30$ lies to the left of μ ; since $5\% < 50\%$, $X = 80$ lies to the right of μ .

**Solution — Step 3 — Read the z values**

Let $Z = -z_2$ and $Z = z_1$ correspond to $X = 30$ and $X = 80$.

$$P(-z_2 < Z < 0) = 0.40$$

$$P(0 < Z < z_1) = 0.45$$

$$\Rightarrow P(0 < Z < z_2) = 0.40$$

$$\Rightarrow \boxed{z_2 = 1.28}$$

$$\Rightarrow \boxed{z_1 = 1.645}$$

(from the normal table)

Solution — Step 4 — Find μ and σ Using $Z = \frac{X - \mu}{\sigma}$:

$$-1.28 = \frac{30 - \mu}{\sigma}$$

$$\boxed{\mu - 1.28\sigma = 30} \quad (1)$$

$$1.645 = \frac{80 - \mu}{\sigma}$$

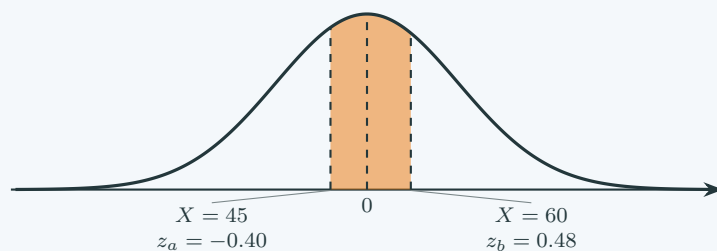
$$\boxed{\mu + 1.645\sigma = 80} \quad (2)$$

(2) - (1):

$$2.925\sigma = 50 \implies \sigma = 17.094 \quad \mu = 30 + 1.28(17.094) = 30 + 21.88 = 51.88$$

Solution — Step 5 — StandardisingSecond division $\Rightarrow P(45 < X < 60)$, with $\mu = 51.88$, $\sigma = 17.094$:

$$z_a = \frac{45 - 51.88}{17.094} = \frac{-6.88}{17.094} = -0.40, \quad z_b = \frac{60 - 51.88}{17.094} = \frac{8.12}{17.094} = 0.48$$

**Solution — Step 6 — Answer**

$$\begin{aligned} P(45 < X < 60) &= P(-0.40 < Z < 0.48) \\ &= P(-0.40 < Z < 0) + P(0 < Z < 0.48) \\ &= P(0 < Z < 0.40) + P(0 < Z < 0.48) \quad (\text{by symmetry}) \\ &= 0.1554 + 0.1844 = 0.3398 \end{aligned}$$

Students placed in the second division = $0.3398 \times 100 \approx 34\%$ **Problem 7.6 — Life of Dry Battery Cells**

A sample of 100 dry battery cells tested to find the length of life produced the following results:

$$\bar{x} = 12 \text{ hours}, \quad \sigma = 3 \text{ hours}$$

Assuming the data to be normally distributed, what percentage of battery cells are expected to have life

(i) more than 15 hours, (ii) less than 6 hours, (iii) between 10 and 14 hours?

(U.P. III Semester, Dec. 2003)

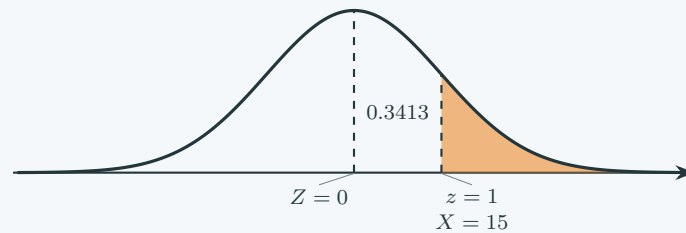
Solution — Step 1 — Data

Let $X \rightarrow$ length of life of a dry battery cell, $X \sim N(\mu, \sigma^2)$ with

$$\mu = \bar{x} = 12 \text{ hours}, \quad \sigma = 3 \text{ hours}, \quad Z = \frac{X - \mu}{\sigma} = \frac{X - 12}{3}.$$

Solution — (i) Life more than 15 hours

$$\text{When } X = 15 : \quad z = \frac{15 - 12}{3} = 1$$

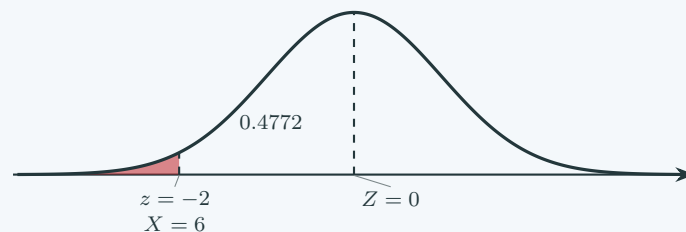


$$P(X > 15) = P(Z > 1) = 0.5 - P(0 < Z < 1) = 0.5 - 0.3413 = 0.1587$$

$\Rightarrow$ **15.87%** of the cells last more than 15 hours.

Solution — (ii) Life less than 6 hours

$$\text{When } X = 6 : \quad z = \frac{6 - 12}{3} = -2$$

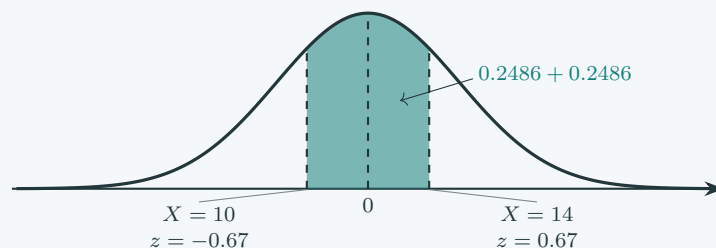


$$P(X < 6) = P(Z < -2) = 0.5 - P(-2 < Z < 0) = 0.5 - P(0 < Z < 2)$$

$$= 0.5 - 0.4772 = 0.0228 \quad \Rightarrow \quad \mathbf{2.28\%}$$

Solution — (iii) Life between 10 and 14 hours

$$X = 10 : z = \frac{10 - 12}{3} = -0.67 \quad \quad X = 14 : z = \frac{14 - 12}{3} = 0.67$$



$$\begin{aligned}
 P(10 < X < 14) &= P(-0.67 < Z < 0.67) \\
 &= P(-0.67 < Z < 0) + P(0 < Z < 0.67) \\
 &= 2P(0 < Z < 0.67) \quad (\text{by symmetry}) \\
 &= 2 \times 0.2486 = 0.4972
 \end{aligned}$$

49.72% of the battery cells have a life between 10 and 14 hours

7.3 Summary of Answers — Normal Distribution

Summary

No.	Problem	Answer
1	Soldiers over 6 feet (out of 2000)	≈ 250
2	Mean and S.D. (31% under 45, 8% over 64)	$\mu \approx 50, \sigma \approx 10$
3	Mean height and S.D. of men	$\mu = 65.42'', \sigma = 3.29''$
4	Lowest mark of A / highest mark of B	83 / 72
5	Students in the second division	$\approx 34\%$
6	Battery life: > 15h / < 6h / 10–14h	15.87% / 2.28% / 49.72%

Consolidated Formula Sheet — Unit I

All Distributions at a Glance

Quantity	Binomial	Poisson	Normal
	$B(n, p)$	$P(\lambda)$	$N(\mu, \sigma^2)$
p.d.f.	${}^nC_x p^x q^{n-x}$	$\frac{e^{-\lambda} \lambda^x}{x!}$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
Type	Discrete	Discrete	Continuous
MGF	$(q + pe^t)^n$	$e^{\lambda(e^t-1)}$	$e^{\mu t + \frac{\sigma^2 t^2}{2}}$
Mean	np	λ	μ
Variance	npq	λ	σ^2
S.D.	$\sqrt{npq}$	$\sqrt{\lambda}$	σ

General Definitions

$$E(X) = \begin{cases} \sum x f(x) \\ \int_{-\infty}^{\infty} x f(x) dx \end{cases} \quad \text{Var}(X) = E(X^2) - (E(X))^2 \quad \sigma = \sqrt{\text{Var}(X)}$$

$$F(x) = P(X \leq x) \quad M_X(t) = E(e^{tX}) \quad \mu'_r = \left\{ \frac{d^r}{dt^r} M_X(t) \right\}_{t=0}$$

$$\text{Var}(X) = \mu'_2 - (\mu'_1)^2 \quad Z = \frac{X - \mu}{\sigma} \sim N(0, 1) \quad \lambda = np \quad (n \text{ large, } p \text{ small})$$

Thank You

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The Uniform Distribution

Definition, Properties and Worked Problems

1. Definition

Definition: Uniform Distribution

A random variable X is said to follow a **uniform distribution** on (a, b) if its probability density function (pdf) has the form

$$f(x) = \begin{cases} k, & a < x < b \\ 0, & \text{elsewhere.} \end{cases}$$

The density is the *same* at every point of the interval, so no value is more likely than any other. Notation: $X \sim U(a, b)$.

Determining the constant k

Since the total probability must equal 1,

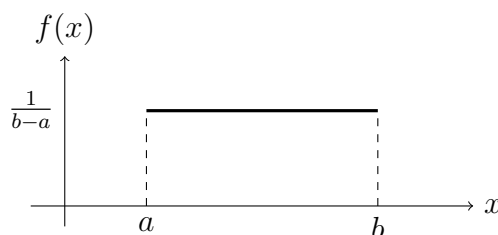
$$\begin{aligned} \int_a^b f(x) dx &= 1 \\ \Rightarrow \int_a^b k dx &= 1 \\ \Rightarrow k [x]_a^b &= 1 \\ \Rightarrow k (b - a) &= 1 \end{aligned}$$

Hence

$$k = \frac{1}{b - a}$$

The pdf of the uniform distribution

$$f(x) = \begin{cases} \frac{1}{b - a}, & a < x < b \\ 0, & \text{elsewhere.} \end{cases}$$



2. Standard Results

Standard Results

For $X \sim U(a, b)$:

1. Mean: $\mu = \frac{a+b}{2}$

2. Variance: $\sigma^2 = \frac{(b-a)^2}{12}$

3. Cumulative distribution function: $F(x) = P(X \leq x) = \frac{x-a}{b-a}, \quad a \leq x \leq b$

4. Probabilities are simply *length ratios*: $P(c < X < d) = \frac{d-c}{b-a}$ for $a \leq c < d \leq b$

3. Worked Problems

Problem 1

If X is uniformly distributed in $[-\alpha, \alpha]$ with $\alpha > 0$, determine α such that $P(X > 1) = \frac{1}{3}$.

Solution

Given $X \sim U[-\alpha, \alpha]$, the pdf of X is

$$f(x) = \frac{1}{\alpha - (-\alpha)} = \frac{1}{2\alpha}, \quad -\alpha < x < \alpha.$$

It is given that $P(X > 1) = \frac{1}{3}$:

$$\int_1^\alpha f(x) dx = \frac{1}{3} \implies \int_1^\alpha \frac{1}{2\alpha} dx = \frac{1}{3}$$

$$\frac{1}{2\alpha} [x]_1^\alpha = \frac{1}{3} \implies \frac{1}{2\alpha} (\alpha - 1) = \frac{1}{3}$$

$$\begin{aligned} \frac{1}{2} - \frac{1}{2\alpha} &= \frac{1}{3} \\ \frac{1}{2\alpha} &= \frac{1}{2} - \frac{1}{3} = \frac{1}{6} \end{aligned}$$

$$\boxed{\alpha = 3}$$

Problem 2

A bus travels between two cities, A and B , which are 100 miles apart. If the bus has a breakdown, the distance X of the point of breakdown from city A has a uniform distribution $U[0, 100]$.

(a) There are service garages in city A , city B , and midway between cities A and B . If a breakdown occurs, a tow truck is sent from the garage closest to the point of breakdown. What is the probability that the tow truck has to travel more than 10 miles to reach the bus?

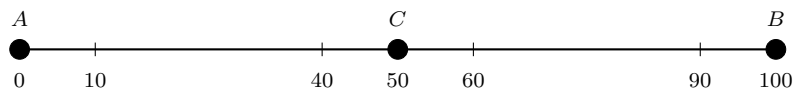
(b) Would it be more “efficient” if the three service garages were placed at 25, 50 and 75 miles from city A ? Explain.

Solution — Part (a)

Given that the distance between cities A and B is 100 miles and $X \sim U[0, 100]$, the pdf of X is

$$f(x) = \frac{1}{100 - 0} = \frac{1}{100}, \quad 0 < x < 100.$$

The garages are located at 0, 50 and 100:



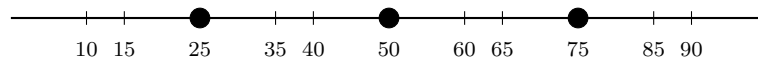
The tow truck travels more than 10 miles precisely when the breakdown occurs in (10, 40) or in (60, 90). Hence

$$\begin{aligned} P(\text{truck travels} > 10) &= P((10 < X < 40) \cup (60 < X < 90)) \\ &= P(10 < X < 40) + P(60 < X < 90) \\ &= \int_{10}^{40} \frac{1}{100} dx + \int_{60}^{90} \frac{1}{100} dx \\ &= \frac{1}{100} [(40 - 10) + (90 - 60)] = \frac{1}{100} [30 + 30] = \frac{60}{100} \end{aligned}$$

$P = 0.6 = 60\%$

Solution — Part (b)

Now the garages are placed at 25, 50 and 75 miles from A :



A tow truck now has to travel more than 10 miles only when the breakdown occurs in one of the four short intervals below:

$$\begin{aligned} P &= P((10 < X < 15) \cup (35 < X < 40) \cup (60 < X < 65) \cup (85 < X < 90)) \\ &= \int_{10}^{15} \frac{dx}{100} + \int_{35}^{40} \frac{dx}{100} + \int_{60}^{65} \frac{dx}{100} + \int_{85}^{90} \frac{dx}{100} \end{aligned}$$

$$= \frac{1}{100} [5 + 5 + 5 + 5] = \frac{20}{100}$$

$$P = 0.2 = 20\%$$

Conclusion

The probability of a long trip falls from 60% to 20%. Placing the garages at 25, 50 and 75 is therefore considerably **more efficient**.

Problem 3

A bus arrives every 10 minutes at a bus stop. Assuming the waiting time X for the bus is uniformly distributed, find the probability that a person has to wait (a) for more than 7 minutes, (b) between 2 and 7 minutes.

Solution

Given: X = waiting time for the bus at the stop, $X \sim U(0, 10)$. Hence

$$f(x) = \frac{1}{10}, \quad 0 < x < 10.$$

(a)

$$P(X > 7) = \int_7^{10} f(x) dx = \int_7^{10} \frac{1}{10} dx = \frac{1}{10} (10 - 7) = \frac{3}{10}$$

$$P(X > 7) = 0.3$$

(b)

$$P(2 < X < 7) = \int_2^7 f(x) dx = \int_2^7 \frac{1}{10} dx = \frac{1}{10} (7 - 2) = \frac{5}{10}$$

$$P(2 < X < 7) = 0.5$$

4. Exercise Problems

Problem 4 (Difficulty: Medium)

Let X be uniformly distributed on the interval $(-2, 3)$.

(a) Find $P(|X| > 1)$.

(b) Find $E(X)$ and $E(X^2)$, and hence verify that $\text{Var}(X) = \frac{(b-a)^2}{12}$.

Solution — Part (a)

Here $a = -2$, $b = 3$, so

$$f(x) = \frac{1}{3 - (-2)} = \frac{1}{5}, \quad -2 < x < 3.$$

The event $|X| > 1$ means $X < -1$ **or** $X > 1$. These are disjoint, so

$$\begin{aligned} P(|X| > 1) &= P(-2 < X < -1) + P(1 < X < 3) \\ &= \int_{-2}^{-1} \frac{1}{5} dx + \int_1^3 \frac{1}{5} dx = \frac{1}{5}[1 + 2] = \frac{3}{5} \end{aligned}$$

$$P(|X| > 1) = \frac{3}{5} = 0.6$$

Solution — Part (b)

$$E(X) = \int_{-2}^3 \frac{x}{5} dx = \frac{1}{5} \left[\frac{x^2}{2} \right]_{-2}^3 = \frac{1}{5} \cdot \frac{9 - 4}{2} = \frac{1}{2}$$

$$E(X^2) = \int_{-2}^3 \frac{x^2}{5} dx = \frac{1}{5} \left[\frac{x^3}{3} \right]_{-2}^3 = \frac{1}{5} \cdot \frac{27 + 8}{3} = \frac{7}{3}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{7}{3} - \frac{1}{4} = \frac{28 - 3}{12} = \frac{25}{12}$$

Check: $\frac{a+b}{2} = \frac{-2+3}{2} = \frac{1}{2} = E(X)$ and $\frac{(b-a)^2}{12} = \frac{(3+2)^2}{12} = \frac{25}{12} = \text{Var}(X)$.
✓

Problem 5 (Difficulty: Medium)

A random variable X is uniformly distributed on $(0, 5)$. Consider the quadratic equation in t

$$4t^2 + 4tX + (X + 2) = 0.$$

- (a) Find the probability that the roots of this equation are real.
- (b) Given that $X > 1$, find the probability that the roots are real.

Solution — Part (a)

Given $X \sim U(0, 5)$, the pdf of X is

$$f(x) = \frac{1}{5 - 0} = \frac{1}{5}, \quad 0 < x < 5.$$

The roots of $4t^2 + 4tX + (X + 2) = 0$ are real if and only if the discriminant is non-negative:

$$\begin{aligned} D &= (4X)^2 - 4(4)(X + 2) \geq 0 \\ 16X^2 - 16X - 32 &\geq 0 \\ X^2 - X - 2 &\geq 0 \\ (X - 2)(X + 1) &\geq 0 \end{aligned}$$

Since $0 < X < 5$, we always have $X + 1 > 0$, so the condition reduces to

$$X - 2 \geq 0 \implies X \geq 2.$$

Therefore

$$P(\text{roots are real}) = P(X \geq 2) = \int_2^5 \frac{1}{5} dx = \frac{1}{5} [x]_2^5 = \frac{5-2}{5}$$

$$P = \frac{3}{5} = 0.6 = 60\%$$

Solution — Part (b)

We require the conditional probability $P(X \geq 2 \mid X > 1)$. Since $\{X \geq 2\} \subset \{X > 1\}$,

$$P(X \geq 2 \mid X > 1) = \frac{P(\{X \geq 2\} \cap \{X > 1\})}{P(X > 1)} = \frac{P(X \geq 2)}{P(X > 1)}.$$

Now

$$P(X > 1) = \int_1^5 \frac{1}{5} dx = \frac{5-1}{5} = \frac{4}{5}, \quad P(X \geq 2) = \frac{3}{5}$$

so that

$$P(X \geq 2 \mid X > 1) = \frac{3/5}{4/5} \implies P = \frac{3}{4} = 0.75$$

5. Summary

Summary of Results

- $X \sim U(a, b)$: $f(x) = \frac{1}{b-a}$ on (a, b) , 0 elsewhere.
- $\mu = \frac{a+b}{2}$, $\sigma^2 = \frac{(b-a)^2}{12}$.
- Probabilities are length ratios: $P(c < X < d) = \frac{d-c}{b-a}$.

- **Problem 1:** $\alpha = 3$.
- **Problem 2:** (a) 0.6 (b) 0.2 — the second placement is more efficient.
- **Problem 3:** (a) 0.3 (b) 0.5.
- **Problem 4:** (a) 0.6 (b) $E(X) = \frac{1}{2}$, $\text{Var}(X) = \frac{25}{12}$.
- **Problem 5:** (a) 0.6 (b) 0.75.